

Dimension Floors for Quotient-Faithful Encoding: Architectural Bounds in Representation Learning

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Abstract

Paper I established that standard predictive and compression criteria cannot certify quotient faithfulness. This paper asks a prior question: for a declared quotient space Q , what latent dimension is geometrically necessary for faithful encoding to be possible at all? The answer is a floor, not a programme: topology-faithful encoding of Q in a Euclidean latent space $Z \subseteq \mathbb{R}^m$ requires $m \geq \text{emb}(Q)$, the minimum Euclidean embedding dimension of Q . Any non-bypassable layer through which every information path must pass is subject to the same floor. These are necessary conditions for declared quotients; they say nothing about learnability, trainability, or quotient discovery. The paper does not provide a general architecture theory for networks with skip connections, attention, or routing mechanisms—such architectures require case-by-case bottleneck analysis beyond what is settled here.

The topological floor is one rung of a hierarchy. Stricter contracts (smooth, bi-Lipschitz, isometric) impose strictly larger floors: the bi-Lipschitz floor at fixed distortion scales with the doubling dimension; the C^∞ isometric floor is $\Theta(n^2)$ for n -manifolds by Nash–Günther; and Euclidean latent space of any finite dimension is not universal for C^∞ isometric representation. Which contract is operative for quotient-faithful learning depends on the objective and remains open.

Within those limits, the paper develops the constructive complement: what dimension suffices for a topology-faithful encoding to exist? The classical embedding hierarchy—Menger–Nöbeling ($\leq 2n + 1$), Whitney ($\leq 2n$ for smooth manifolds), Haeffliger savings for connected quotients—gives tight or near-tight existence bounds. For covering dimension at most three, the Thurston–Perelman geometrisation classification yields a complete finite lookup table. For product-structured quotients, tight bounds are derived: $\text{emb}(\prod S^{n_i}) = 1 + \sum n_i$, a strict improvement over the naive sum. A quantitative C^1 approximation-to-embedding bridge (Hirsch 1976; Federer–NSW) relates these existence bounds to smooth neural approximants. For quotients of non-integer box-counting dimension, the Sauer–Yorke–Casdagli extension gives a prevalence sufficiency condition $m > 2 d_{\text{box}}$, illustrated for the Lorenz attractor ($d_{\text{box}} \approx 2.062 \Rightarrow m \geq 5$).

These results are existence bounds, not discovery guarantees. A network of exactly the right width can still converge to a traversal solution, a graded-compression solution, or a metrically distorted solution. The dimension floor characterises one class of geometric impossibility; the discovery question is an open problem (Paper 2 [Farkas, 2026b], §9; this paper, §12.3).

About this series

A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. The series *Quotient Faithfulness in Representation Learning* develops a unified account of when and why this gap arises and what is required to close it. A representation is quotient-faithful if it maps semantically equivalent inputs to the same code and faithfully encodes the space of semantically distinct states as a topological structure in the latent space. Three structural manifestations of criterion silence prevent standard training from achieving this: the embedding floor, graded compression, and quotient-respect failure.

Paper I [Farkas, 2026a]: Establishes that predictive loss does not certify quotient faithfulness; develops the embedding-space program and the three structural manifestations.

Paper II [Farkas, 2026b]: Identifies recovery routes and admissibility sources. Establishes the graded-compression structure and its Kolmogorov complexity foundations.

Paper III [Farkas, 2026c] (this paper): Derives geometric lower and constructive upper bounds on the latent space dimension required for topology-faithful encoding in the single-quotient setting.

Paper IV [Farkas, 2026d]: Extends the dimension-floor programme to region-varying quotient structure via quotient atlases. Develops the given-partition, learned-partition, and emergent-discovery tiers.

Paper V [Farkas, 2026e]: Maps the framework against representation-learning traditions and examines fourteen bodies of experimental evidence through the embedding-space lens.

Paper VI [Farkas, 2026f]: Shows that partial structural knowledge — generator specifications — constrains the admissible quotient class and imposes certifiable architecture floors without naming the equivalence relation explicitly. Introduces the Generator-Constrained Discovery (GCD) scheme.

Paper VII [Farkas, 2026g]: Empirical validation on benchmark dynamical systems (phase cylinder, Hopf fibration, double pendulum) and an industrial case study (Color Fantasy ship-hotel energy system). Benchmarks against Hamiltonian Neural Networks [Greydanus et al., 2019].

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Notation

Symbol	Meaning
$\mathcal{X}$	Observation space
$\sim$	Admissible nuisance relation
$Q = \mathcal{X}/\sim$	Quotient (semantic state) space
$q : \mathcal{X} \rightarrow Q$	Quotient map
$f : \mathcal{X} \rightarrow Y$	Prediction target
$t : Q \rightarrow Y$	Task map ($f = t \circ q$)
$E : \mathcal{X} \rightarrow \mathcal{Z}$	Encoder (representation map)
$\mathcal{Z}$	Latent space
$\tilde{E} : Q \rightarrow \mathcal{Z}$	Induced quotient representation
E_{label}	Label representation: $E_{\text{label}}(x) = f(x)$
$\dim(Q)$	Covering dimension of Q
$\text{emb}(Q)$	Minimum Euclidean embedding dimension of Q
$\tau(\phi(Q))$	Reach of the embedded submanifold $\phi(Q) \subset \mathbb{R}^m$
r_c	Graded quotient kernel $r_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$
$d_c = 1 - r_c$	Semantic ultrapseudometric
$Q_{c,\varepsilon} = \mathcal{X}/\sim_{c,\varepsilon}$	Hard sub-quotient at scale ε
$\text{emb}(Q)$ $2 \dim(Q)$	$\leq$ Whitney upper bound (smooth manifolds)
$\text{emb}(Q)$ $2 \dim(Q) + 1$	$\leq$ Menger–Nöbeling upper bound (general compact metric)
$\sqrt{3/5} \tau$	NSW homological stability threshold ($\approx 0.7746\tau$)
$d_{\text{box}}(A)$	Box-counting dimension of compact set A (SYC extension, §5.2)

1 Introduction

Position in the series. A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. Paper I of this series [Farkas, 2026a] proved that three structural traps prevent standard criteria from certifying quotient faithfulness. Paper 2 [Farkas, 2026b] identified the sources through which the correct equivalence structure can enter learning and the mechanisms that escape the traversal and graded compressions. Both papers presuppose that the architecture is at least *capable* of faithful encoding. This paper addresses that presupposition. Paper 5 [Farkas, 2026e] situates the dimension floor and the upper bound programme within the broader theoretical landscape.

This paper. This paper addresses a single question: given a declared quotient space Q , what latent width and depth does an encoder need to represent Q faithfully?

The question arises as the constructive complement to two prior papers. Farkas [2026a] proved that standard training criteria — samplewise accuracy, sufficient statistics, compression objectives — cannot certify quotient faithfulness. Three independent failure modes exist, each arising from a different way predictive fit and structural faithfulness come apart. Farkas [2026b] characterised when recovery occurs despite those failures, via admissibility sources and stochastic escape. Both papers presuppose that the architecture is at least *capable* of faithful encoding. The present paper asks when that presupposition is satisfied.

The answer has two parts. The lower bound is settled by classical topology: any topology-faithful encoding of Q requires the final latent space Z to have $\dim(Z) \geq \text{emb}(Q)$, the minimum Euclidean embedding dimension of Q . This is a constraint on where the quotient must ultimately live, not a per-layer requirement. No optimisation innovation can circumvent it. The upper bound — what dimension *suffices* for Z , and what architecture achieves it — is the constructive question.

Prerequisites. Section 2 provides a self-contained two-page summary of the framework and the three failure modes from Farkas [2026a]. No prior reading of Farkas [2026b] is required for the architecture results, though §4 draws on the graded quotient kernel developed there.

Running example. A cyclic dynamical process with observation space $C = S^1 \times [0, 1]$ and declared quotient $q : C \rightarrow S^1$. The circle S^1 is the simplest compact non-contractible space and the canonical test case: $\dim(S^1) = 1$, $\text{emb}(S^1) = 2$. Any encoder whose final latent space has $\dim(Z) = 1$ provably cannot represent S^1 faithfully. The minimum sufficient final-latent-space width is 2, achievable constructively (Guliyev–Ismailov 2016, §6).

The gap this paper addresses. The series of which this paper is part is a theory of the gap between *task-relative adequacy* and *lawful abstraction*. Paper I established three ways that gap arises; Paper 2 characterised three routes to closing it. This paper addresses a prior geometric condition: the dimension floor determines when the gap is even bridgeable by architecture. A network whose final latent space is too small cannot faithfully encode the declared quotient regardless of training procedure, objective, or inductive bias. The floor precedes all three routes.

Structure. Part I (§§1–2) sets up notation and restates the three failure modes. Part II (§§3–4) establishes lower bounds. Part III (§§5–10) develops the upper bound programme in full, including the Sauer–Yorke–Casdagli extension to fractal compact sets (§5.2). Part IV (§§11–12) presents the empirical validation and open problems.

2 Background: Quotient Spaces, the Three Failure Modes, and the Width Floor

This section is self-contained. Readers familiar with [Farkas \[2026a\]](#) may proceed to §3.

2.1 Setting and Definitions

Definition 2.1 (Admissible nuisance relation).

An **admissible nuisance relation** $\sim$ on $\mathcal{X}$ identifies observations that differ only in semantically irrelevant ways. Its equivalence classes are **nuisance fibres**.

Definition 2.2 (Lawful quotient, encoder, topology-faithful encoding).

The **lawful quotient** is (Q, q) where $Q = \mathcal{X}/\sim$ and $q : \mathcal{X} \rightarrow Q$ is the canonical projection. An encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ induces $\tilde{E} : Q \rightarrow \mathcal{Z}$ with $E = \tilde{E} \circ q$ when it is quotient-respecting. $\tilde{E}$ is **topology-faithful** if it is a homeomorphism onto its image. The prediction target $f : \mathcal{X} \rightarrow \mathcal{Y}$ factors as $f = t \circ q$ for a **task map** $t : Q \rightarrow \mathcal{Y}$.

The key distinction: a representation can achieve perfect predictive accuracy (task sufficiency) while the induced $\tilde{E}$ is not injective (faithfulness fails). The three theorems below establish how and why this happens.

2.2 The Three Failure Modes

Theorem 4.6 — Finite-Sample Traversal [[Farkas, 2026a](#)]

For any finite training set $D \subset \mathcal{X}$ and any target values, there exists a one-dimensional assignment $\tau : D \rightarrow [0, 1]$ and readout h with $h \circ \tau = f|_D$. Samplewise supervision cannot distinguish a quotient-faithful representation from a one-dimensional traversal of the sample cloud. (In Paper I’s revised framing, this is the $k = 1$ case of graded compression.)

Theorem 4.4 — Task-Map Collapse [[Farkas, 2026a](#)]

The label representation $E_{\text{label}}(x) = f(x)$ is the unique $\preceq$ -minimum among all sufficient representations. The map it induces on Q is the task map t . If t is non-injective, E_{label} does not embed Q .

Theorem 3.2, Lemma 3.4, Corollary 3.6 — Embedding-Floor Obstructions [[Farkas, 2026a](#)]

- (3.2) If $\tilde{E} : Q \rightarrow \mathbb{R}^m$ is topology-faithful, then $m \geq (Q)$.
- (3.4) If $\tilde{E} : Q \rightarrow \mathbb{R}^m$ is topology-faithful, then Q embeds homeomorphically in $\mathbb{R}^m$.
- (3.6) If $m < (Q)$, no continuous map $\tilde{E} : Q \rightarrow \mathbb{R}^m$ can be topology-faithful.

2.3 The Width Floor as Architectural Constraint

Corollary 3.6 implies more than a dimension bound. Since $\dim(Q) \leq \text{emb}(Q)$ always (any compact metric space of covering dimension n embeds in $\mathbb{R}^{2n+1}$ by Menger–Nöbeling), the tight floor is:

Architectural width floor — scope and limits.

Primary Claim (no structural assumption on architecture). The final latent space Z must have $\dim(Z) \geq \text{emb}(Q)$. This is a statement about the target space of the induced map $\tilde{E} : Q \rightarrow Z$, not about individual layers.

Secondary Claim (architecture-level, per-path). For any directed path in the computation graph from input $\mathcal{X}$ to the final latent Z , every non-bypassable layer on that path must have width $\geq \text{emb}(Q)$. For chain architectures this reduces to $\min_{\ell} w_{\ell} \geq \text{emb}(Q)$. For residual, attention, or multi-branch architectures the condition is per-path: quotient-faithful encoding requires the existence of at least one non-bypassable path whose every layer has width $\geq \text{emb}(Q)$, not that every individual path does.

What the floor does not say. It does not require every layer to achieve width $\geq \text{emb}(Q)$, and does not assert that any single intermediate layer independently encodes Q faithfully. Quotient faithfulness is a property of the full encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$. Farkas [2026a] make this explicit in Definition 2.4 and §3 (around Theorem 3.2): Theorem 3.3 of that paper (new numbering) constrains only the exposed latent space Z , not any internal layer, intermediate activation, or architectural bottleneck inside the network that computes the encoder. The Secondary Claim above constrains the computation graph’s non-bypassable structure, not internal layer representations in isolation. Analysing the bypass structure of a specific architecture (residual streams, attention heads, gating mechanisms, mixture-of-experts routing) is a graph-theoretic question that this paper treats only in the chain case.

Width is the only continuous capacity. The floor above is activation-agnostic. Three observations clarify its content:

1. *Continuous-coordinate argument.* For a non-bypassable layer ℓ of width w_{ℓ} on a directed path, the induced map $h^{(\ell)} : Q \rightarrow \mathbb{R}^{w_{\ell}}$ has image $h^{(\ell)}(Q) \subseteq \mathbb{R}^{w_{\ell}}$. A topology-faithful encoder requires $h^{(\ell)}|_Q$ to be injective, hence a topological embedding, hence $\text{emb}(Q) \leq w_{\ell}$. This holds regardless of whether the activation σ is smooth, piecewise-linear, or otherwise continuous.
2. *Depth, connectivity, and piecewise-structure richness.* Stacking fully-connected layers multiplies piecewise-approximation complexity — polyhedral cell count for ReLU (Zaslavsky-bounded in terms of widths and input dimension), oscillation capacity for smooth activations, compositional chart count generally. This increases the number of affine or smooth charts available to approximate a curved $\tilde{E}$ inside the w_{ℓ} -dimensional budget. It does not alter the floor: a union of pieces in $\mathbb{R}^{w_{\ell}}$ is still a subset of $\mathbb{R}^{w_{\ell}}$, covering-dimension $\leq w_{\ell}$. Combinatorial cell count does not supplement continuous embedding dimension.
3. *Geometric character of the image (ReLU refinement).* For ReLU activations, $h^{(\ell)}(Q) \subseteq \mathbb{R}_{\geq 0}^{w_{\ell}}$ decomposes by activation pattern into a union of affine pieces glued along their shared lower-dimensional faces. The topological-embedding conclusion is unchanged; what changes is that the image is a continuous-piecewise-affine object rather than a smooth submanifold. Claims about “layer ℓ embeds Q ” in this setting mean embedding into the continuous-piecewise-affine image, not into a smooth manifold.

Terminology note. We use “non-bypassable layer” for the graph-theoretic condition “every directed path on which this layer sits crosses it, with no bypass within the path.”

The earlier literature sometimes uses “bottleneck” for this, but “bottleneck” in neural-network usage commonly refers to a deliberate narrow latent in an autoencoder-style wide-narrow-wide architecture. The two notions are distinct: autoencoder bottlenecks are specific architectural choices, while non-bypassable layers are a graph-theoretic property satisfied automatically by every layer of a plain feedforward encoder. We reserve “non-bypassable layer” for the graph property throughout.

The full remediation framework has four mechanisms, each matched to one failure mode [Farkas, 2026b]:

Failure mode	Trap type	Theorem	Remedy
Traversal ambiguity	Finite-sample artefact	Theorem 4.6	Stochasticity
Task-map collapse	Population-level optimum	Theorem 4.4	Structural regularisation or population selection
Dimension obstruction	Hard geometric constraint	Corollary 3.6	Architecture redesign

Architecture is the precondition: the dimension floor must be satisfied before the other mechanisms can operate. Stochasticity addresses only the finite-sample traversal; it is neither necessary nor sufficient for full quotient recovery. The graded compression requires either a structural prior that reshapes the loss landscape (equivariant architecture, JEPA-style noise, topology loss) or population-level selection across networks with a multi-dimensional fitness criterion. Population selection is introduced as a fourth recovery mechanism in Farkas [2026b], §5.9; it is capable of escaping the graded compression that single-network stochasticity cannot. The present paper concerns only the architecture precondition.

2.4 Stronger Contracts, Stronger Floors

The floor of §2.3 is a *topological* floor: it is the minimum dimension at which Q can be embedded as a topological subspace of Euclidean latent space. For learning purposes, a stronger contract is typically required. A representation that preserves only topology may distort distances or angles arbitrarily, breaking downstream metric-dependent operations (nearest-neighbour retrieval, Lipschitz classifiers, perturbation stability, transfer). This subsection states the floors that stricter contracts impose, and observes that each can substantially exceed the topological floor.

The hierarchy of contracts. For a compact metric space (Q, d_Q) and latent $Z \subseteq \mathbb{R}^m$ with Euclidean metric d_Z , the following conditions on $\tilde{E} : Q \rightarrow Z$ form a strict hierarchy:

Contract	Condition on $\tilde{E}$	Floor
Topological	Homeomorphism onto its image	$m \geq \text{emb}(Q)$ (Prop. 3.1)
C^∞ smooth	C^∞ embedding (Q a manifold)	$m \geq 2 \dim(Q)$ (Whitney)
Bi-Lipschitz, distortion D	$D^{-1}d_Q(q, q') \leq d_Z(\tilde{E}q, \tilde{E}q') \leq Dd_Q(q, q')$	$m \geq \text{emb}_{\text{bi-Lip}}(Q, D)$ (Prop. 2.4)
C^1 isometric	C^1 isometric embedding	$m \geq \dim(Q) + 1$ (Nash–Kuiper)
C^∞ isometric	C^∞ isometric embedding	$m \geq \frac{(n+2)(n+5)}{2}$ for n -manifold (Nash–Günther)

Each stricter contract implies the weaker ones, so the topological floor is a lower bound on every floor above. The converse need not hold: for specific Q , the metric floor can

strictly exceed the topological one.

Proposition 3.2 (Bi-Lipschitz floor, parametric).

Let (Q, d_Q) be a compact metric space and let $D \geq 1$. Define

$$\text{emb}_{\text{bi-Lip}}(Q, D) := \min\{m : Q \text{ admits a bi-Lipschitz embedding into } \mathbb{R}^m \text{ with distortion } \leq D\},$$

with the convention that $\text{emb}_{\text{bi-Lip}}(Q, D) = \infty$ if no such embedding exists for any finite m .

Let $\tilde{E} : Q \rightarrow Z \subseteq \mathbb{R}^m$ be bi-Lipschitz with distortion $\leq D$. Then:

- (i) $m \geq \text{emb}_{\text{bi-Lip}}(Q, D)$.
- (ii) $\text{emb}_{\text{bi-Lip}}(Q, D) \geq \text{emb}(Q)$ for all $D \geq 1$.
- (iii) (Doubling special case.) If Q has doubling dimension $\dim_{\text{doub}}(Q) = \delta < \infty$, then there exist $D_0(\delta) < \infty$ and $C(\delta) < \infty$ such that $\text{emb}_{\text{bi-Lip}}(Q, D) \leq C(\delta)$ for all $D \geq D_0(\delta)$ (Naor–Neiman 2012). Conversely, $\text{emb}_{\text{bi-Lip}}(Q, D) \geq c \cdot \delta$ for all finite D , where $c > 0$ is an absolute constant (volume-comparison lower bound, classical).

Proof sketch. (i) holds by definition. (ii) A bi-Lipschitz embedding is *a fortiori* a topological embedding, and the minimum dimension for topological embedding is $\text{emb}(Q)$. (iii) The upper bound is Theorem 1.2 of Naor and Neiman [2012]. The lower bound is a volume-comparison argument: a bi-Lipschitz image in $\mathbb{R}^m$ can contain at most $(CD)^m$ disjoint balls of a given radius relative to the ambient diameter, while Q with doubling dimension δ requires at least $c_1 \cdot 2^{c_2 \delta}$ such balls for some $c_1, c_2 > 0$. Matching these gives $m \gtrsim \delta$. $\square$

Theorem 3.3 (Smooth and isometric floors, classical; stated without proof).

Let Q be a compact Riemannian n -manifold.

- (a) *Whitney embedding.* Q admits a smooth embedding into $\mathbb{R}^{2n+1}$, and generically into $\mathbb{R}^{2n}$ (Whitney).
- (b) *Nash–Kuiper C^1 isometric.* For $m \geq n + 1$, any short C^∞ embedding $Q \hookrightarrow \mathbb{R}^m$ can be C^0 -approximated by a C^1 isometric embedding [Nash, 1954, Kuiper, 1955].
- (c) *Nash–Günther C^∞ isometric.* For $m \geq \max(n + 5, \frac{(n+2)(n+5)}{2})$, Q admits a C^∞ isometric embedding into $\mathbb{R}^m$ [Nash, 1956, Günther, 1991]. Subsequent work has tightened this bound.

Implications. Three points follow:

1. *The topology floor is a lower bound, not the operative one.* For Q a smooth n -manifold, the topology floor is $\text{emb}(Q) \in [n, 2n + 1]$. The bi-Lipschitz floor at fixed distortion can reach the doubling dimension (for fractal or highly curved Q , substantially above n). The C^∞ isometric floor is $\Theta(n^2)$, quadratically larger. A representation that satisfies Prop. 3.1 with equality may distort metric structure arbitrarily.
2. *The right contract for learning is an empirical question.* No existing theorem characterises which contract is operative for a given training objective and downstream

task. Cross-entropy with softmax is sensitive to angular structure but not scale; contrastive objectives are explicitly metric; regression is Lipschitz-sensitive at a quantitative level set by the output loss. In each case the relevant floor is likely above the topological one, but by how much depends on the objective.

3. *Stronger contracts can exceed finite-dimensional Euclidean reach.* The flat torus admits no C^2 isometric embedding into $\mathbb{R}^3$ (classical), a C^1 isometric embedding into $\mathbb{R}^3$ (Nash–Kuiper), and a C^∞ isometric embedding into some $\mathbb{R}^m$ with $m \geq 4$. For general Riemannian manifolds the C^∞ isometric floor grows at least quadratically and can be larger in construction-specific cases. Euclidean latent space of any finite dimension is not universal for isometric representation. The question of which contract is the operative floor for quotient-faithful learning is taken up as an open problem in §12.6.

Part II — Lower Bounds

From embedding theory and graded geometry

3 Lower Bounds from Embedding Theory

3.1 The Architectural Reading of the Embedding Obstruction

Corollary 3.6 of Farkas [2026a] is usually stated as a result about the induced map $\tilde{E} : Q \rightarrow Z$. Its architectural reading is stronger.

Proposition 3.1 (Width floor for the latent space and non-bypassable layers).

Let $E : \mathcal{X} \rightarrow \mathcal{Z}$ be an encoder with final latent space $\mathcal{Z} \cong \mathbb{R}^m$.

- (i) **Latent space floor.** If $\tilde{E} : Q \rightarrow \mathcal{Z}$ is topology-faithful, then $m \geq \text{emb}(Q)$.
- (ii) **Non-bypassable layer floor.** Suppose layer ℓ is a *non-bypassable layer*: every directed path from $\mathcal{X}$ to $\mathcal{Z}$ in the computation graph passes through it, with no bypass. If $\tilde{E} : Q \rightarrow \mathcal{Z}$ is topology-faithful, then $w_\ell \geq \text{emb}(Q)$.

Proof. (i) follows immediately from Lemma 3.4 of Farkas [2026a]: a topology-faithful $\tilde{E}$ is a topological embedding of Q in $\mathcal{Z} \cong \mathbb{R}^m$, so $\text{emb}(Q) \leq m$.

(ii) The non-bypassable condition forces $\tilde{E}$ to factor as $Q \xrightarrow{h} \mathbb{R}^{w_\ell} \xrightarrow{g} \mathcal{Z}$ where h is the induced map at layer ℓ and g is the continuation. Since $\tilde{E} = g \circ h$ is injective, h must be injective: if $h(q_1) = h(q_2)$ then $\tilde{E}(q_1) = g(h(q_1)) = g(h(q_2)) = \tilde{E}(q_2)$, contradicting injectivity. An injective continuous map from compact Q to Hausdorff $\mathbb{R}^{w_\ell}$ is a homeomorphism onto its image — a topological embedding. Therefore Q embeds homeomorphically in $\mathbb{R}^{w_\ell}$, forcing $w_\ell \geq \text{emb}(Q)$. $\square$

Scope. The floor $\text{emb}(Q)$ is stated for a single global quotient Q . The generalisation to a *quotient atlas* — regionally varying quotient structure with per-region floors plus region-recognition and gluing-coherence costs — is developed in Paper 4 (forthcoming), and an Open Problem entry appears in §12.5 below.

Remark 3.1.1 (What this does and does not imply for architecture design).

Part (i) is the primary architectural consequence: the *final* latent space Z must have $\dim(Z) \geq \text{emb}(Q)$. This requires no assumptions about the internal architecture.

Part (ii) extends this to intermediate layers, but only under the non-bypassable condition. A layer is non-bypassable only if there is literally no directed path in the computation graph that routes around it — a condition satisfied by vanilla feed-forward encoders but not by residual networks, attention-based architectures, or multi-branch encoders.

The floor does *not* require every layer to have width $\geq \text{emb}(Q)$. In a network with a wide embedding layer, a narrow intermediate layer, and a wide subsequent projection, only the narrow non-bypassable layer triggers the analysis in (ii). Non-bypassable layers can be arbitrarily narrow in architectures that route around them; layers that

are bypassable (not on every path) are not constrained by (ii) at all.

Quotient faithfulness is a property of the *full encoder* $E : \mathcal{X} \rightarrow \mathcal{Z}$, not of any individual layer. The bounds above constrain where topology-faithful encoding is geometrically impossible, not where it is guaranteed. The upper bound programme of Part III establishes where it is possible.

Remark 3.1.2 (What sub-floor encoders actually produce: entanglement, not destruction).

Proposition 3.1 establishes a floor: topology-faithful encoding of Q in $\mathbb{R}^m$ requires $m \geq \text{emb}(Q)$. Below the floor, no topology-faithful map exists. But a trained encoder still produces some continuous map $\tilde{E} : Q \rightarrow Z$; the theorem says this map cannot be injective in a topologically stable way, not that it encodes nothing about Q .

The correct characterisation of the sub-floor outcome is *entanglement*. Any continuous map from Q into an insufficiently large Z must identify some distinct points of Q — it folds Q ’s topology into the available dimensions, creating false adjacencies and collapsed distinctions. Information about Q ’s structure is present in Z , but it is co-mingled with whatever else occupies those dimensions. Semantic distinctions that cannot be separated geometrically in Z become entangled with other encoded content.

The claim is not that a sub-floor encoder produces a useless representation — partial information about Q may still be recoverable. The claim is that no readout from Z can recover the full topology of Q independently of whatever else Z encodes. Resolving the entanglement requires either increasing $\dim(Z)$ to meet the floor, or accepting that Q ’s structure can only be decoded jointly with the other content in Z .

In Kolmogorov complexity terms [Dingle and Hutter, 2026], the dimension floor is a *feasibility* constraint rather than a complexity-difficulty constraint. The issue is not that topology-faithful encoders in insufficient dimension have high KC; it is that they do not exist. No amount of simplicity bias, algorithmic probability sampling, or search efficiency can find a solution that is geometrically impossible. This is precisely where the KC framework reaches its limit: it explains selection among feasible solutions, but the dimension floor removes solutions from the feasible set entirely.

Remark 3.1.3 (The missing architecture-to-embedding bridge).

Proposition 3.1 handles two clean cases: the final latent space and non-bypassable layers in chain architectures. For the general case — deep networks with skip connections, multi-head attention, gating mechanisms, or mixture-of-experts routing — no analogue of (ii) is proved here. Such architectures may route different parts of the input through different capacity paths, and whether any sub-path contains a non-bypassable layer for the quotient-relevant information depends on both the architecture and the data distribution.

A full architecture-to-embedding bridge would characterise, for a general computation graph $\mathcal{G}$ and a declared quotient Q , the minimum widths at each node of $\mathcal{G}$ that are necessary and sufficient for the full encoder to admit a topology-faithful induced map. This is an open problem. The bounds in this paper are necessary conditions; their sufficiency for architectures beyond the strict chain case is not established.

Remark 3.1.4 (Practical floors for the final latent space).

Applying Proposition 3.1(i) to common quotients: for $Q = S^1$, $\dim(Z) \geq 2$; for any closed orientable surface $(S^2, \mathbb{T}^2, \Sigma_g)$, $\dim(Z) \geq 3$; for $SO(3) \cong \mathbb{R}P^3$, $\dim(Z) \geq 5$; for the eight Thurston geometries in dimension 3, $\dim(Z) \geq 4$ (with Wall’s upper bound ≤ 5); for $\mathbb{T}^n = (S^1)^n$, $\dim(Z) \geq n + 1$. These are hard floors on the final latent space Z . They are not per-layer requirements.

3.2 Process-Generated Quotients Have Additional Structure

The quotient spaces arising from real data-generating processes are not arbitrary topological spaces. They carry regularity that purely abstract manifolds do not.

Symmetry quotients arise when the admissible nuisance relation is a group action. The quotient $Q = \mathcal{X}/G$ is a homogeneous space with known geometry whenever G is a compact Lie group. The embedding dimension is determined by the representation theory of G and is known in closed form for the classical groups.

Smooth constraint manifolds arise from ODE/DAE systems, mechanical joints, molecular conformation spaces, and pose estimation problems. The manifold inherits smooth structure from the dynamics; the relevant geometric invariants are computable from the constraint equations.

Low-dimensional attractors of dissipative systems are compact, often smooth, and have effective dimension far below the ambient state space. The gap between ambient and intrinsic dimension is large, and the architectural requirements are correspondingly milder than the ambient dimensionality suggests.

Stratified spaces from switching dynamics or contact mechanics carry process-derived regularity: strata are smooth manifolds of varying dimension, and the incidence structure between strata is constrained by the dynamics.

For all four classes, the process structure supplies not only the topology of Q but information about its geometric type — symmetry group, Thurston geometry, product decomposition — that tightens the architecture bounds beyond the generic Menger–Nöbeling fallback.

Corollary 3.2 (Product decomposition lower bound).

If $Q = Q_1 \times \cdots \times Q_k$ where each Q_i is a compact metric space with $\dim(Q_i) = d_i$, then any topology-faithful encoding requires latent width $m \geq \sum_i d_i$.

Proof. By the product dimension formula for compact metric spaces [Engelking, 1978], $\dim(Q) = \sum_i d_i$. Corollary 3.6 then gives $m \geq \dim(Q)$. $\square$

4 Scale-Indexed Architecture Bounds

When admissibility is graded rather than hard, the architecture floor inherits a scale parameter. This section cites the graded quotient kernel developed in Farkas [2026b]; readers may treat the cited theorems as black boxes without loss of continuity.

4.1 Graded Quotients and the Semantic Ultrapseudometric

Definition 4.1 (Graded quotient filtration; Farkas 2026b, Def. 3.2).

A **graded quotient kernel** $r_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ is reflexive, symmetric, and min-transitive. Setting $d_c = 1 - r_c$ yields a **semantic ultrapseudometric**. For each $\varepsilon \in [0, 1]$, define $x \sim_{c,\varepsilon} x' \iff d_c(x, x') \leq \varepsilon$ and the **hard sub-quotient** $Q_{c,\varepsilon} = \mathcal{X} / \sim_{c,\varepsilon}$. These nest: $\varepsilon_1 < \varepsilon_2$ implies Q_{c,ε_1} is a finer quotient than Q_{c,ε_2} .

Theorem 3.5 — Three equivalent formulations of graded admissibility [Farkas, 2026b]

For $r_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$, the following are equivalent: (i) r_c is a graded quotient kernel; (ii) $d_c = 1 - r_c$ is a semantic ultrapseudometric; (iii) $\{Q_{c,\varepsilon}\}_{\varepsilon \in [0,1]}$ is a nested filtration of quotients recovering r_c via $r_c(x, x') = 1 - d_c(x, x')$.

Theorem 3.7 — Scale-by-scale certification [Farkas, 2026b]

If d_c can be verified at scale ε , then the hard sub-quotient $Q_{c,\varepsilon}$ is certified in latent space up to distortions $\phi_c(\varepsilon)$ and $\psi_c^{-1}(\varepsilon)$. The hard-case limit $\varepsilon \rightarrow 0$ recovers exact quotient faithfulness.

4.2 Architecture Bounds Scale with Semantic Resolution

The embedding obstruction applies at every level of the filtration.

Proposition 4.2 (Scale-indexed width floor).

For any $\varepsilon \in [0, 1]$, a topology-faithful encoding of the hard sub-quotient $Q_{c,\varepsilon}$ requires latent width at least $\text{emb}(Q_{c,\varepsilon})$. The width floor is non-decreasing as $\varepsilon \rightarrow 0$: finer semantic resolution demands at least as much latent width.

Proof. Apply Proposition 3.1 to $Q_{c,\varepsilon}$ at each ε . Monotonicity follows because $\varepsilon_1 < \varepsilon_2$ implies Q_{c,ε_1} is a topologically richer quotient with $\text{emb}(Q_{c,\varepsilon_1}) \geq \text{emb}(Q_{c,\varepsilon_2})$ generically. $\square$

Remark 4.2.1 (Practical consequence for approximate quotient encoding).

A practitioner who can only verify admissibility at scale $\varepsilon > 0$ still faces a hard width floor: the architecture must be wide enough to embed $Q_{c,\varepsilon}$. Coarser semantic resolution (ε large) permits narrower encoders; exact quotient recovery ($\varepsilon \rightarrow 0$) imposes the full $\text{emb}(Q)$ floor. This makes explicit that the width requirement is not binary but scales continuously with the precision of the admissibility contract.

Part III — The Upper Bound Programme

What width suffices for faithful encoding to exist?

Scope of Part III.

The results in this part answer the **existence** question: what is the most compressed latent space in which a topology-faithful map *can* exist? They say nothing about whether a network of that width will *discover* a topology-faithful encoding through gradient descent on any particular objective and data distribution. They say nothing about whether a Euclidean latent is the appropriate ambient space. They say nothing about whether a given neural architecture realises the required embedding class. Existence of the map and learnability of the map are logically independent: a network could have exactly the right width and still converge to a traversal solution, a graded-compression solution, or a topologically correct but metrically distorted solution that training cannot distinguish from faithful. The discovery question is addressed as an open problem in §12.3.

5 The Embedding Hierarchy

Three classical theorems stratify the upper bound problem by the structural assumptions one is willing to make about Q . For quick reference, Table 1 summarises the minimum latent dimension requirements before the detailed proofs.

Quotient type	Min. width	latent	Reason
S^1 (phase / cycle)	≥ 2		scalar bottleneck provably fails
$\mathbb{T}^2 = S^1 \times S^1$ (two independent phases)	≥ 3		two independent periods; each factor contributes 1 to dim
S^2 / closed orientable surface	≥ 3		Jordan–Brouwer
$SO(3)$	≥ 5		dimension 3, parallelisable; Hopf 1940
Thurston 3-geometries	4–5		Perelman classification; 8 families; see Table 4
Compact n -manifold (general)	$\geq n + 1$		invariance of domain
$Q_1 \times Q_2 \times \cdots \times Q_k$ (product)	$\geq \sum_i \dim(Q_i)$		product dimension formula + Theorem 3.3

Table 1: Minimum latent width requirements for topology-faithful encoding. Hard floors, not heuristics.

5.1 Menger–Nöbeling, Whitney, and Haeffliger

Theorem (Menger–Nöbeling, 1931–1932 [Menger, 1932, Nöbeling, 1931]).

Every separable metrizable space of covering dimension at most n embeds homeomorphically in $\mathbb{R}^{2n+1}$.

This is the universal fallback, applying to arbitrary compact metric quotients with no smoothness assumption. The bound $2n + 1$ is tight: the n -skeleton of the $(2n + 2)$ -

simplex is a compact n -dimensional polyhedron that cannot embed in $\mathbb{R}^{2n}$ [Flores, 1935, van Kampen, 1932]. The proof is non-constructive (Baire category).

Theorem (Whitney Strong Embedding, 1944 [Whitney, 1944]).

Every smooth n -dimensional manifold (Hausdorff, second-countable) embeds smoothly in $\mathbb{R}^{2n}$.

The improvement from $2n + 1$ to $2n$ exploits smooth structure via the Whitney trick: generic immersions $M^n \hookrightarrow \mathbb{R}^{2n}$ produce double points with opposite intersection signs, cancelled by embedded Whitney discs. The bound $2n$ is tight when n is a power of 2: $\mathbb{R}P^{2^k}$ cannot embed in $\mathbb{R}^{2^{k+1}-1}$, proved via dual Stiefel–Whitney class obstructions. When n is not a power of 2, the Haefliger–Hirsch theorem improves this to $2n - 1$.

Theorem (Haefliger, 1961 [Haefliger, 1961]).

Let M^n be a compact, k -connected smooth manifold with $n \geq 2k + 3$. Then M embeds smoothly in $\mathbb{R}^{2n-k}$.

Each unit of connectivity saves one ambient dimension. For simply-connected manifolds ($k = 1$) with $n \geq 5$, embedding in $\mathbb{R}^{2n-1}$ is guaranteed. The condition $n \geq 2k + 3$ is sharp. The architectural implication: a quotient Q with $\pi_1(Q) = 0$ and $\dim(Q) = n \geq 5$ has $\text{emb}(Q) \leq 2n - 1$, reducing the required width by one.

5.2 Fractal Compact Sets: The Sauer–Yorke–Casdagli Extension

The three theorems above assume either compact metric spaces (Menger–Nöbeling) or smooth manifolds (Whitney, Haefliger). Real data-generating processes produce quotient spaces that may be neither: chaotic attractors are fractal, having non-integer Hausdorff and box-counting dimension and no tangent plane at almost every point. The standard dimension floor analysis does not directly apply. The following theorem provides the appropriate extension.

Theorem (Sauer, Yorke, and Casdagli, 1991 [Sauer et al., 1991] — Embedology).

Let $A \subset \mathbb{R}^k$ be compact with box-counting dimension $d_{\text{box}}(A)$. If $m > 2d_{\text{box}}(A)$, then for *prevalent* (measure-theoretically almost every) Lipschitz maps $F : \mathbb{R}^k \rightarrow \mathbb{R}^m$, F is injective on A . Moreover, F is an embedding on every smooth manifold contained within A .

Prevalence is the infinite-dimensional analogue of Lebesgue-measure-zero: a property holds prevalently if the set of maps failing it has measure zero under every compactly supported measure on the space of Lipschitz maps. The condition $m > 2d_{\text{box}}$ is tight in the sense that the set of self-intersecting maps has positive measure when $m \leq 2d_{\text{box}}$.

Contrast with the obstruction results of this paper. Paper III’s floor theorems (§§3–4) are *deterministic obstructions*: they show that *no* continuous injective map $Q \hookrightarrow \mathbb{R}^m$ exists when $m < \text{emb}(Q)$, for specific smooth quotients Q where $\text{emb}(Q)$ is computed from topological covering dimension and characteristic class obstructions. The SYC theorem is a *prevalence sufficiency* result: it shows that *almost every* Lipschitz map is injective when $m > 2d_{\text{box}}$, without asserting that any particular map works or that $m = 2d_{\text{box}}$ is the correct floor. The two results are complementary:

	Paper III (smooth)	SYC (fractal)
Object	Compact smooth manifold	Compact set in $\mathbb{R}^k$
Dimension used	Covering dimension n	Box-counting d_{box}
Floor	$\text{emb}(Q) \geq n + 1$ (tight)	$m > 2 d_{\text{box}}$
Type of result	Deterministic obstruction	Prevalent sufficiency
Map quantifier	No map works below	Almost every map works above

Architectural implication for fractal quotient spaces. For a chaotic attractor A with box-counting dimension d_{box} , the SYC theorem implies that an encoder whose latent space has $m > 2 d_{\text{box}}$ will, with probability one over random initialisations (in the appropriate measure), find an injective representation. Encoders with $m \leq 2 d_{\text{box}}$ will, with positive probability, produce representations with self-intersections regardless of training procedure and data size.

For the Lorenz attractor ($d_{\text{box}} \approx 2.062$): $2 d_{\text{box}} \approx 4.124$, so the SYC condition requires $m \geq 5$. This is an architectural recommendation, not a proved lower bound. Whether a deterministic lower bound $\text{emb}(\text{Lorenz}) \geq 5$ holds — i.e. whether *every* continuous injective map requires $m \geq 5$ — is an open question stated in §12.4.

Remark 4.3.1 (SYC and the quotient faithfulness framework).

The SYC theorem was developed for the time-series reconstruction problem (Takens’ delay embedding), not for representation learning. Its hypothesis is that A is a fixed compact set and maps are drawn randomly; in learning, the map is optimised while the set is unknown. The connection to the present framework is structural: the condition $m > 2 d_{\text{box}}$ plays the same role as $m \geq \text{emb}(Q)$ — it separates the architecturally feasible from the architecturally infeasible. Extending the SYC prevalence argument to the learning setting (where the map is not random but gradient-descent-selected) is an open problem; see §12.4.

5.3 Tight Bounds for Specific Quotients and Characteristic Class Obstructions

The non-embedding lower bounds in Table 2 follow from characteristic class obstructions and classical topological arguments. If M^n embeds in $\mathbb{R}^{n+c}$ (codimension c), Whitney duality gives $w(TM) \cdot w(\nu) = 1$ in $H^*(M; \mathbb{Z}/2)$, where ν is the normal bundle of rank c . Therefore the dual Stiefel–Whitney class $\bar{w}(M) = w(TM)^{-1}$ satisfies $\bar{w}_i(M) = 0$ for $i > c$; equivalently, $\bar{w}_k(M) \neq 0$ forces $\text{emb}(M^n) \geq n + k$.

For $\mathbb{R}P^2$: in $H^*(\mathbb{R}P^2; \mathbb{Z}/2) \cong \mathbb{Z}/2[a]/(a^3)$, $w(T\mathbb{R}P^2) = (1 + a)^3 = 1 + a + a^2$ over $\mathbb{Z}/2$ (since $\binom{3}{1} = \binom{3}{2} = 3 \equiv 1 \pmod{2}$). Inverting: $\bar{w}(\mathbb{R}P^2) = 1 + a$, so $\bar{w}_1 = a \neq 0$. Stiefel–Whitney theory gives $\text{emb} \geq 3$; the correct lower bound $\text{emb} \geq 4$ follows from non-orientability: by Jordan–Brouwer, any closed surface embedded in $\mathbb{R}^3$ must be orientable.

For $\mathbb{R}P^3 \cong SO(3)$: as a compact Lie group, $SO(3)$ is parallelisable, so $w(TSO(3)) = 1$ and $\bar{w} = 1$ — all dual Stiefel–Whitney classes vanish. There is no Stiefel–Whitney obstruction. The lower bound $\text{emb}(\mathbb{R}P^3) \geq 5$ is Hopf’s 1940 result [Hopf, 1940], proved via linking number arguments; see also Bredon [1993], Theorem VI.4.2 for a modern treatment. The upper bound ≤ 5 follows from Wall’s theorem [Wall, 1965].

Quotient Q	$\dim(Q)$	$\text{emb}(Q)$	Status	Key fact
$[a, b]$ (interval)	1	1	Tight	Contractible
S^1 (circle)	1	2	Tight	$S^1 \not\hookrightarrow \mathbb{R}^1$: invariance of domain
S^n (n -sphere)	n	$n + 1$	Tight	Closed n -mfd $\not\hookrightarrow \mathbb{R}^n$; standard embedding
$\mathbb{T}^n = (S^1)^n$	n	$n + 1$	Tight	Inductive revolution: $\mathbb{T}^n \subset \mathbb{R}^{n+1}$
Σ_g (genus- g surface)	2	3	Tight	Orientable; algebraic construction in $\mathbb{R}^3$
$\mathbb{R}P^2$	2	4	Tight	Non-orientable; Jordan–Brouwer forces $\mathbb{R}^4$
$\mathbb{R}P^3 \cong SO(3)$	3	5	Tight	Hopf 1940: $\mathbb{R}P^3 \not\hookrightarrow \mathbb{R}^4$; Wall: $\leq \mathbb{R}^5$
S^3 (3-sphere)	3	4	Tight	$S^3 \subset \mathbb{R}^4$; closed 3-mfd $\not\hookrightarrow \mathbb{R}^3$
$\mathbb{T}^3$ (3-torus)	3	4	Tight	Inductive revolution
$S^m \times S^n$	$m + n$	$m + n + 1$	Tight	Revolution; closed $(m + n)$ -mfd obstruction
$\mathbb{C}P^2$	4	7	Tight	$\sigma(\mathbb{C}P^2) = 1 \Rightarrow$ no smooth emb. in $\mathbb{R}^6$ (signature); Cappell–Shaneson $\leq \mathbb{R}^7$

Table 2: Minimum Euclidean embedding dimensions for quotients arising in representation learning.

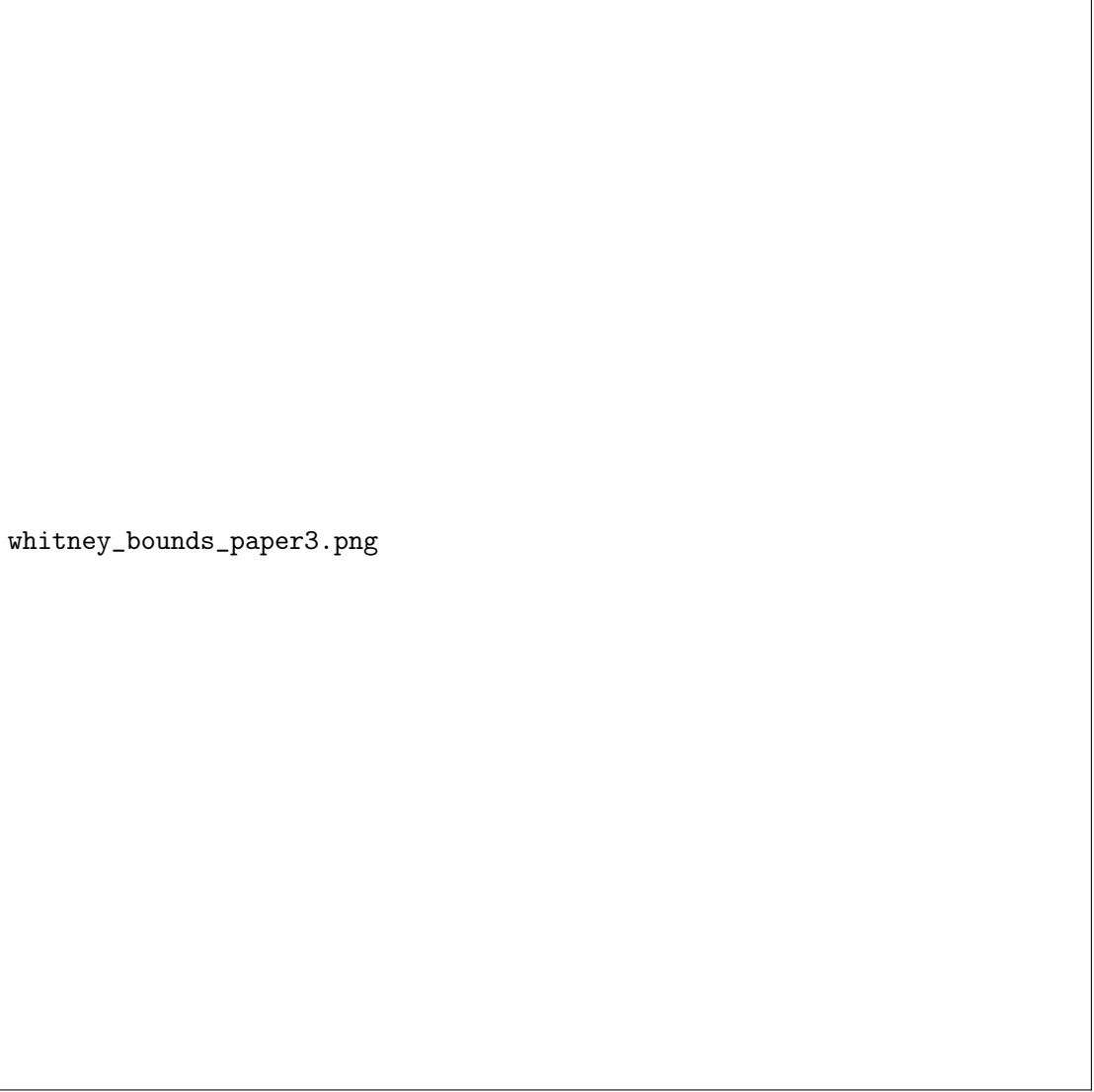
For $\mathbb{C}P^2$: the Stiefel–Whitney approach does not give the correct lower bound here, and would be wrong to invoke without correction. One can compute $w(T\mathbb{C}P^2) = (1 + h)^3 = 1 + h + h^2$ in $H^*(\mathbb{C}P^2; \mathbb{Z}/2) \cong \mathbb{Z}/2[h]/(h^3)$ (where $|h| = 2$), giving $\bar{w}(\mathbb{C}P^2) = 1 + h$, so $\bar{w}_2 = h \neq 0$. By Whitney duality this yields only $\text{emb} \geq n + k = 4 + 2 = 6$, i.e. codimension ≥ 2 — *not* codimension ≥ 3 .

The tight lower bound $\text{emb}(\mathbb{C}P^2) \geq 7$ follows from a signature obstruction. A classical result in 4-manifold topology states that any closed oriented smooth 4-manifold admitting a smooth embedding in $\mathbb{R}^6$ must have signature $\sigma = 0$. (This is a consequence of the Whitney self-intersection pairing: a smooth codimension-2 embedding of a closed oriented 4-manifold in $\mathbb{R}^6$ bounds a Seifert hypersurface whose intersection form must be metabolic, forcing $\sigma = 0$; see e.g. [Freedman and Quinn \[1990\]](#).) Since $\sigma(\mathbb{C}P^2) = 1 \neq 0$, no smooth embedding $\mathbb{C}P^2 \hookrightarrow \mathbb{R}^6$ exists. Hence $\text{emb}(\mathbb{C}P^2) \geq 7$. The upper bound $\text{emb}(\mathbb{C}P^2) \leq 7$ is due to Cappell and Shaneson [[Cappell and Shaneson, 1974](#)], who give an explicit smooth embedding in $\mathbb{R}^7$.

Figure 1 summarises the embedding bounds for all specific quotients treated in this paper against the generic Whitney $2n + 1$ bound, showing where Haefliger connectivity, the product formula, and signature obstructions tighten the floor below Whitney.

Remark 4.5.2 (Contract-level floors enforced architecturally).

The distinction between topological, smooth, and isometric floors in the hierarchy of §2.4 is not purely theoretical. E(3)-equivariant architectures for quantum-materials discovery [[Hassani-Vasmejani et al., 2026](#)] enforce exact equivariance under rotation and translation — an isometric-contract condition — via explicit Wigner-D matrices in message-passing layers. Each tensor order ℓ the network supports contributes the irreducible $SO(3)$ representation of dimension $2\ell + 1$, which is already the minimal faithful representation after full symmetry reduction (spherical harmonics are symmetric and traceless). A network supporting all orders 0 through ℓ therefore requires



whitney_bounds_paper3.png

Figure 1: Embedding dimension lower bounds for specific quotients covered in this paper. Blue bars: intrinsic covering dimension n . Red bars: tight lower bound $\text{emb}(Q)$, derived from Proposition 3.1 and the characteristic class obstructions of §5.3. Dashed line: generic Whitney $2n + 1$ upper bound for arbitrary compact metric spaces. The gap between red and dashed reflects topological savings not captured by dimension alone: Haefliger’s connectivity theorem (S^3, T^2), the product formula (T^2), and the signature obstruction ($\mathbb{C}P^2$). Figure generated by `whitney_bounds_figure.py` in the companion code repository.

latent dimension at least $(\ell + 1)^2$. For a network supporting up to $\ell = 2$ (sufficient for most molecular force fields), this is 9 dimensions, exceeding the topological embedding floor of $\mathbb{R}P^3 \cong SO(3)$ at 5. This is a concrete current setting where the isometric-contract floor and the topological-contract floor genuinely diverge.

6 One-Dimensional Quotients: The Guliyev–Ismailov Hierarchy

For $\dim(Q) = 1$, the lower bound is $\text{emb}(Q) = 1$ (contractible Q) or $\text{emb}(Q) = 2$ ($Q = S^1$). The upper bound question reduces to univariate function approximation. The unifying framework is Ismailov’s ridge-function theory [Ismailov, 2020]: every single-neuron computation is a ridge function $g(\mathbf{w} \cdot x - \theta)$, and ridge-function composition theory governs what one-layer and two-layer networks can and cannot represent. The Guliyev–Ismailov results give the tightest possible constructive prescriptions.

Theorem (Guliyev–Ismailov, 2016 [Guliyev and Ismailov, 2016]).

A single hidden layer feedforward network (SLFN) with a *single neuron* in the hidden layer, equipped with an algorithmically constructible C^∞ sigmoidal activation, can approximate any continuous function on a compact subset of $\mathbb{R}$ to arbitrary precision. The construction terminates in finitely many steps.

Applied to $Q = S^1$: the topology-faithful encoding $\tilde{E} : S^1 \rightarrow \mathbb{R}^2$ writes as $\theta \mapsto (f_1(\theta), f_2(\theta))$ where each $f_i : [0, 2\pi) \rightarrow \mathbb{R}$ is periodic. A single-neuron hidden layer per coordinate is the minimal upper bound, matching the lower bound from Corollary 3.6 exactly.

Theorem (Guliyev–Ismailov, 2017 [Guliyev and Ismailov, 2017]).

An SLFN with hidden weights fixed at 1 and exactly *two neurons* can approximate any continuous function on a compact subset of $\mathbb{R}$. The same fixed-weight architecture *cannot* approximate all continuous multivariate functions, regardless of the number of neurons.

The negative result is the function-approximation analogue of Corollary 3.6: just as no topology-faithful encoding of a k -dimensional quotient exists into $\mathbb{R}^{k-1}$, no fixed-weight SLFN universally approximates multivariate functions. The dimension barrier appears at both levels.

Theorem (Guliyev–Ismailov, 2021 [Guliyev and Ismailov, 2021]).

A two hidden layer feedforward network (TLFN) with weights fixed as the unit coordinate vectors of $\mathbb{R}^d$ and $3d + 2$ total hidden neurons can approximate any continuous d -variable function to arbitrary precision.

For a d -dimensional quotient, a topology-faithful encoding requires $m = \text{emb}(Q)$ coordinate functions each of d variables, achievable with $3d + 2$ neurons per output. For $d = 1$: five total hidden neurons is the tight constructive bound.

Remark 6.1 (Practical consequence for 1D quotients).

For any process-generated quotient of covering dimension 1 — cyclic dynamics, periodic signals, scalar phase variables — the architecture is settled. Minimum latent width is 2 (for $Q = S^1$) or 1 (for $Q = [a, b]$). A single hidden layer with one neuron per coordinate (trainable weights) or two neurons per coordinate (fixed weights) suffices. These are not asymptotic existence results: the proofs provide explicit algorithms for the activation function at any desired precision.

Architecture	Weights	Neurons	Capability	Source
SLFN	Trainable	1	Univariate universal (constructive)	G–I 2016
SLFN	Fixed (= 1)	2	Univariate universal; multivariate impossible	G–I 2017
SLFN	Fixed	Any finite	Cannot universally approximate multivariate	G–I 2017
TLFN	Fixed (unit vectors)	$3d + 2$	d -variable universal (constructive)	G–I 2021

Table 3: The Guliyev–Ismailov hierarchy for 1D quotient encoding.

Remark 6.2 (Extension to non-Euclidean input spaces [Ismailov, 2026a]).

Ismailov [2026a] extend both shallow and deep-narrow universality theorems to networks whose inputs range over an arbitrary topological space X . The first affine layer is replaced by a finite family of continuous *feature maps* $f_i \in \mathcal{A}(X)$, playing the role of inner products $w \cdot x$ in the Euclidean case. Two structural conditions on $\mathcal{A}(X)$ are identified: the D-property (density of univariate superpositions, sufficient for arbitrary-width universality) and the finite-dimensional composition property (embedding of compact subsets of X into $\mathbb{R}^n$, sufficient for deep-narrow universality with explicit width bounds). For products of compact metric spaces $X = \prod_p X_p$ with $\sum_p \dim(X_p) = M$, the Ostrand extension of the Kolmogorov superposition theorem yields width bounds $\leq 2M + m + 3$ for m -output networks. For a process-generated quotient $Q = \prod_i Q_i$ with $\dim(Q) = \sum_i \dim(Q_i)$, this extends the G–I hierarchy from $\mathbb{R}^d$ -valued encoders to quotients with non-Euclidean geometry. A companion result [Ismailov, 2026c,b] establishes the same principle for locally convex spaces and extends the Chen–Chen operator approximation theorem to branch-trunk (DeepONet-type) architectures with non-Euclidean inputs.

7 The C^1 Approximation-to-Embedding Bridge

The Guliyev–Ismailov results concern sup-norm approximation of real-valued functions; topology-faithful encoding requires a homeomorphism $\tilde{E} : Q \rightarrow Z$. The gap between these has a precise characterisation.

7.1 Why C^0 Closeness Does Not Suffice

A direct compactness argument shows that a sufficiently good C^0 approximation of a homeomorphic embedding preserves injectivity only if the error is less than half the minimum separation in the image. But C^0 -closeness alone does not guarantee that the approximant is an embedding: the set of C^0 -injective maps is *not* open in $C^0(M, \mathbb{R}^m)$. Concrete counterexample: $\phi(t) = t$ on $[0, 1]$ is injective, but $f_n(t) = t + (1/n)\sin(n^2t)$ satisfies $\|f_n - \phi\|_\infty \rightarrow 0$ while f_n fails to be injective for $n \geq 2$.

7.2 Hirsch’s Theorem: Openness of C^1 Embeddings

Theorem (Hirsch, 1976 [Hirsch, 1976]).

Let M, N be smooth manifolds with M compact. The space $\text{Emb}^k(M, N)$ of C^k embeddings is open in the C^k mapping space $C^k(M, N)$ for every $k \geq 1$.

Applied to the bridge: if $\phi : Q \hookrightarrow \mathbb{R}^m$ is a smooth embedding and the neural encoder f satisfies $\|f - \phi\|_{C^1} < \varepsilon^*$ (maximum of sup-norm error and sup-norm Jacobian error), then f is also a smooth embedding. The threshold ε^* depends on ϕ but not on f .

7.3 The Reach Theorem: Quantifying the Tolerance

Definition 7.3.1 (Federer reach, 1959 [Federer, 1959]).

The **reach** $\tau(\phi(Q))$ of a compact submanifold $\phi(Q) \subset \mathbb{R}^m$ is the largest r such that every point within distance r from $\phi(Q)$ has a unique nearest point on $\phi(Q)$. The reach simultaneously controls local failure (Jacobian degeneration) and global failure (distant parts of Q mapped too close).

Theorem (Niyogi–Smale–Weinberger, 2008 [Niyogi et al., 2008]).

Let $\phi(Q) \subset \mathbb{R}^m$ be a compact submanifold with reach τ . Any compact set A with Hausdorff distance $d_H(A, \phi(Q)) < \varepsilon \leq \sqrt{3/5}\tau \approx 0.7746\tau$ deformation-retracts to $\phi(Q)$ and therefore has the same homotopy type.

As long as the neural encoder f satisfies $\|f - \phi\|_\infty < \sqrt{3/5}\tau(\phi(Q))$, the encoded image $f(Q)$ is homotopy-equivalent to Q . Combining with Hirsch: if additionally $\|Jf - J\phi\|_\infty < \varepsilon^*$, then f is a homeomorphism onto its image.

7.4 Three-Step Architectural Verification Procedure

Step 1. Existence. Confirm $\text{emb}(Q) \leq m$ using the embedding hierarchy of §5. For smooth n -manifolds, Whitney gives $m = 2n$; Haefliger gives $m = 2n - k$ for k -connected Q ($n \geq 2k + 3$).

Step 2. Target map. Fix an explicit homeomorphic embedding $\phi : Q \hookrightarrow \mathbb{R}^m$ using algebraic or geometric constructions. For spheres and tori this is the revolution

embedding; for surfaces the genus- g algebraic construction; for Q with known Thurston geometry, the geometric model.

Step 3. Refinement. Apply the Guliyev–Ismailov result coordinatewise to obtain a neural encoder f with $\|f - \phi\|_{C^1} < \varepsilon^*(\phi)$, where ε^* is governed by the reach $\tau(\phi(Q))$. The required activation function and neuron count are computable from the G–I proofs.

Remark 7.4.1 (Activation smoothness requirement).

The C^1 closeness of Step 3 holds only when the activation function is C^1 and approximation is controlled in the C^1 topology. Standard C^0 universal approximation theorems are insufficient. The G–I constructive activations are C^∞ , satisfying this requirement by design. Non-smooth activations (ReLU, ELU) do not provide C^1 approximation and cannot be used directly in Step 3.

8 Smooth Compact Manifolds up to Dimension 3

For $\dim(Q) \geq 2$, the upper bound programme requires knowing the topological type of Q . The programme has finite scope for $\dim(Q) \leq 3$ because the classification of compact manifolds is complete in these dimensions.

8.1 Dimension 2: Surface Classification

Every compact orientable surface is homeomorphic to S^2 , a torus $\mathbb{T}^2$, or a connected sum of $g \geq 2$ tori (genus- g surface Σ_g). Every non-orientable surface is homeomorphic to a connected sum of copies of $\mathbb{R}P^2$. All compact orientable surfaces embed in $\mathbb{R}^3$; non-orientable surfaces require at least $\mathbb{R}^4$ by Jordan–Brouwer. The upper bound for any 2D quotient requires knowing only orientability and genus, both in principle detectable from the structure of the data-generating process.

8.2 Dimension 3: Thurston–Perelman Geometrisation

Theorem (Thurston, proved by Perelman 2002–2003; [Kleiner and Lott 2008](#); accessible account: [Tao 2006](#)).

Every closed orientable 3-manifold admits a prime decomposition, and each prime piece carries one of eight model geometries: S^3 , $\mathbb{R}^3$, $\mathbb{H}^3$, $S^2 \times \mathbb{R}$, $\mathbb{H}^2 \times \mathbb{R}$, Nil, Sol, $\widetilde{\text{SL}}(2, \mathbb{R})$.

The Poincaré conjecture — that every simply-connected closed 3-manifold is homeomorphic to S^3 — is the spherical special case. Before Perelman, it was unknown whether 3-manifolds could be undecidably complex. His proof resolves this: the classification is finite and complete.

Geometry	emb(Q)	Upper bound	Lower bound
Spherical (S^3)	4	$S^3 \subset \mathbb{R}^4$ (revolution)	Closed 3-mfd $\not\hookrightarrow \mathbb{R}^3$
Euclidean ($\mathbb{T}^3$)	4	$\mathbb{T}^3 \subset \mathbb{R}^4$ (revolution)	$\mathbb{T}^3 \not\hookrightarrow \mathbb{R}^3$: would imply $\mathbb{T}^3 \cong \partial(\text{solid torus})$
$\mathbb{H}^3$	≥ 5	Wall [Wall, 1965]: compact 3-mfd $\hookrightarrow \mathbb{R}^5$	$\bar{w}_2 \neq 0$ for many arithmetic Γ ; codim ≥ 2
$S^2 \times \mathbb{R}$ ($\rightarrow S^2 \times S^1$)	4	$S^2 \times S^1 \subset \mathbb{R}^4$ (revolution)	Closed 3-mfd $\not\hookrightarrow \mathbb{R}^3$
$\mathbb{H}^2 \times \mathbb{R}$ (Seifert fibred)	4–5	Wall [Wall, 1965] $\leq \mathbb{R}^5$	Dim. obstruction ≥ 4 ; tight lower bound open
Nil (Heisenberg)	4–5	Wall [Wall, 1965] $\leq \mathbb{R}^5$	Dim. obstruction ≥ 4 ; tight lower bound open
Sol (Anosov)	4–5	Wall [Wall, 1965] $\leq \mathbb{R}^5$	Dim. obstruction ≥ 4 ; tight lower bound open
$\widetilde{\text{SL}}(2, \mathbb{R})$	4–5	Wall [Wall, 1965] $\leq \mathbb{R}^5$	Dim. obstruction ≥ 4 ; tight lower bound open

Table 4: Minimum latent width for the eight Thurston geometries, with rigorous justification. $\mathbb{H}^3$: Stiefel–Whitney obstructions for arithmetic hyperbolic manifolds push the floor to ≥ 5 . For Nil, Sol, $\mathbb{H}^2 \times \mathbb{R}$, $\widetilde{\text{SL}}(2, \mathbb{R})$: the dimension obstruction gives ≥ 4 and Wall gives ≤ 5 ; the tight value is open. All entries ≤ 5 in the upper bound column are safe architectural prescriptions, not proved tight lower bounds.

The upper bound $\mathbb{R}^5$ for all eight families follows from Wall’s theorem [[Wall, 1965](#)]: every compact connected smooth 3-manifold embeds in $\mathbb{R}^5$, proved using parallelisability

of all closed orientable 3-manifolds [Stiefel, 1936] (so $w_1 = w_2 = 0$) combined with a general-position argument.

Remark 8.2.1 (What the Poincaré conjecture means for the architecture programme, and why it matters for physics-derived quotients).

The motivation for developing the upper bound programme through the Thurston geometries is not to display mathematical machinery for its own sake. A surprisingly large class of real physical systems produces quotients of exactly the 3-dimensional form.

The physical observation. In classical mechanics and field theory, the state space is often $Q = Q_{\text{space}} \times Q_{\text{time}}$, where Q_{space} is a compact smooth 3-manifold (encoding spatial structure, gauge degrees of freedom, or configuration moduli) and Q_{time} is S^1 or $\mathbb{R}$ (encoding temporal phase). Gauge theories on compact spatial slices, rigid-body dynamics ($Q_{\text{space}} = SO(3) \cong \mathbb{R}P^3$), and molecular conformation spaces are all instances. The product structure is exact, not approximate: the spatial and temporal quotients factor because the physics separates.

Why Perelman matters. The simplest nontrivial simply-connected compact 3-manifold is S^3 , which embeds in $\mathbb{R}^4$. Before Perelman, it was an open question whether simply-connected closed 3-manifolds beyond S^3 existed, and more broadly whether the topological complexity of compact 3-manifolds was decidable. His proof resolves both: every compact 3-manifold is either S^3 or decomposes into prime pieces each carrying one of the eight Thurston geometries. The classification is finite and complete. There is no undecidable “wild type”.

Architectural consequence. For any physical system whose spatial quotient Q_{space} is a compact 3-manifold, the required latent width is computable from the Thurston geometry of its prime decomposition:

- Spherical geometry (S^3 , $SO(3) \cong \mathbb{R}P^3$, lens spaces): $\text{emb}(Q_{\text{space}}) \leq 5$; combined with $Q_{\text{time}} = S^1$ via the product rule, $\text{emb}(Q) \leq 6$.
- Euclidean geometry ($\mathbb{T}^3$ for toroidal compactifications): $\text{emb}(Q) \leq 5$ by revolution.
- Hyperbolic, Nil, Sol, or mixed geometries: $\text{emb}(Q_{\text{space}}) \leq 5$ (Wall 1965), giving $\text{emb}(Q) \leq 6$ with the temporal factor.

In every case the latent-width upper bound is finite and small — at most 6 for a system with 3D spatial quotient and 1D temporal phase. This is a concrete architectural prescription, not an asymptotic existence statement.

8.3 Dimension ≥ 4

No complete topological classification of compact manifolds exists in dimension ≥ 4 . The upper bound question must be approached via the specific structure of the process-generating mechanism — symmetry groups, connectivity arguments enabling Haefliger savings, or degree-by-degree analysis — rather than via a finite classification table.

9 Product Quotient Spaces

Modern representation learning frequently encounters product quotients $Q = Q_1 \times \cdots \times Q_k$ where components have different covering dimensions $d_i = \dim(Q_i)$ and embedding dimensions $m_i = \text{emb}(Q_i)$.

9.1 Covering Dimension: Additivity for Manifolds

For compact manifolds (and compact polyhedra), covering dimension is strictly additive: $\dim(M \times N) = \dim(M) + \dim(N)$. This additivity can fail for pathological compact metric spaces [Pontryagin, 1930]: there exist 2-dimensional compacta Π_p with $\dim(\Pi_p \times \Pi_q) = 3 < 4$. The failure goes downward only — $\dim(X \times Y) \leq \dim(X) + \dim(Y)$ holds universally [Hurewicz, 1935]. Cohomological dimension with field coefficients is always strictly additive. For the compact manifold quotients arising in representation learning, dimension additivity holds without caveat.

9.2 The Sphere-Product Formula

Proposition 9.2.1 (Additive upper bound for product embeddings).

If Q_i embeds homeomorphically in $\mathbb{R}^{m_i}$ for each i , then $Q_1 \times \cdots \times Q_k$ embeds in $\mathbb{R}^{m_1 + \cdots + m_k}$, i.e.

$$\text{emb}(Q_1 \times \cdots \times Q_k) \leq \sum_i \text{emb}(Q_i).$$

The C^1 bridge applies coordinatewise: each component ϕ_i is approximated independently.

The additive bound is safe but not tight. The tight result for sphere products is:

Theorem (Tight embedding of sphere products).

For $n_1, n_2, \dots, n_k \geq 1$:

$$\text{emb}(S^{n_1} \times S^{n_2} \times \cdots \times S^{n_k}) = 1 + n_1 + n_2 + \cdots + n_k.$$

Upper bound: the revolution construction extends inductively. Lower bound: the product is a closed $(\sum n_i)$ -manifold, hence does not embed in $\mathbb{R}^{\sum n_i}$ by invariance of domain.

The savings relative to the additive bound $\sum(n_i + 1) = k + \sum n_i$ are exactly $k - 1$: each additional sphere factor saves one ambient dimension beyond the first.

9.3 General Composition Rule and Sub-Encoder Decomposition

For a product $Q = Q_1 \times \cdots \times Q_k$ of compact manifolds:

$$\max_i m_i \leq \text{emb}(Q) \leq \min\left(\sum_i m_i, 2 \sum_i d_i\right).$$

The lower bound is the projection lower bound (the product contains each factor as a retract). The upper bound takes the minimum of the additive product embedding and Whitney applied to the product manifold of dimension $\sum d_i$. For k -connected factors satisfying Haefliger’s condition, a further saving of $\min_i(k_i)$ is available.

Product Q	$\sum_i \text{emb}(Q_i)$	$\text{emb}(Q)$	Saving	Construction
$S^1 \times S^1 = \mathbb{T}^2$	$2 + 2 = 4$	3	1	Revolve $S^1 \subset \mathbb{R}^2$ around axis in $\mathbb{R}^3$
$(S^1)^n = \mathbb{T}^n$	$2n$	$n + 1$	$n - 1$	Inductive revolution: $\mathbb{T}^n \subset \mathbb{R}^{n+1}$
$S^m \times S^n$ ($m, n \geq 1$)	$(m + 1) + (n + 1)$	$m + n + 1$	1	$(x, y) \mapsto ((2 + y_1) \cdot x, y_2, \dots, y_{n+1})$
$S^1 \times S^2$	$2 + 3 = 5$	4	1	Product revolution in $\mathbb{R}^4$
$\mathbb{T}^2 \times S^1 = \mathbb{T}^3$	$3 + 2 = 5$	4	1	Inductive revolution
General manifold product	$\sum \text{emb}(Q_i)$	$\leq 2 \sum d_i$ (Whitney)	variable	Whitney on the product manifold

Table 5: Embedding-dimension savings for product quotients.

The joint encoder decomposes naturally as $E = (E_1, \dots, E_k)$ with $E_i : Q_i \rightarrow \mathbb{R}^{m_i}$. Each sub-encoder is designed according to the prescription for Q_i independently. Example: for $Q = S^1 \times \mathbb{T}^2 \times S^3$, the additive bound gives $m = 2 + 3 + 4 = 9$; the revolution formula gives $m = 1 + (1 + 2 + 3) = 7$; three independent sub-encoders with depths 1, 2, 2 and hidden neuron counts 2, 8, 11 respectively suffice.

9.4 The Kolmogorov–Arnold Connection

The function-approximation side of the product upper bound is governed by the Kolmogorov–Arnold representation theorem [Kolmogorov, 1957, Arnold, 1957]: every continuous $f : [0, 1]^n \rightarrow \mathbb{R}$ writes as

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^n \psi_{q,p}(x_p) \right),$$

where the inner functions $\psi_{q,p}$ are universal and the outer functions Φ_q are f -dependent. Ostrand’s generalisation [Ostrand, 1971] extends this to compact metric spaces of total covering dimension n . The TLFN with $3d + 2$ neurons is the direct neural realisation of this structure. Crucially, K-A universality and topology-faithful embedding are independent properties: when Q has non-trivial topology, fixing the target embedding ϕ first and then applying K-A coordinatewise is required.

10 Architecture Lookup Table

Quotient Q	dim	emb	Architecture (constructive)
$[a, b]$	1	1	SLFN, 1 neuron, smooth G–I activation
S^1	1	2	SLFN, 1 neuron per coord., smooth G–I activation
$\mathbb{T}^2, S^2, \Sigma_g \ (g \geq 0)$	2	3	TLFN, $3(2) + 2 = 8$ neurons, C^1 approx. of revolution / algebraic embedding
$\mathbb{R}P^2$	2	4	TLFN, 8 neurons; width 4 (not 3)
$S^3, S^2 \times S^1, \mathbb{T}^3$	3	4	TLFN, $3(3) + 2 = 11$ neurons, revolution / product embedding
$SO(3) \cong \mathbb{R}P^3$	3	5	TLFN, 11 neurons, width 5; Wall $\leq \mathbb{R}^5$
$\mathbb{H}^3/\Gamma$, Nil, Sol, $\widetilde{\text{SL}}$ manifolds	3	≤ 5	TLFN, 11 neurons, width 5 (Wall upper bound; tight value open)
$S^{n_1} \times \dots \times S^{n_k}$	$\sum n_i$	$1 + \sum n_i$	Independent sub-encoders per factor; revolution construction
$Q_1 \times \dots \times Q_k$ (gen-eral)	$\sum d_i$	$\leq \min(\sum m_i, \sum d_i)$	Independent sub-encoders; width from component rule

Table 6: Minimum sufficient latent width and constructive architecture for topology-faithful encoding. All lower bounds are hard. Upper bounds are from the constructive G–I results and embedding hierarchies.

Remark 10.1 (Symmetry quotients).

For symmetry quotients ($Q = \mathcal{X}/G$ for a compact Lie group G), the upper bound on architecture *form* is resolved by [Kondor and Trivedi \[2018\]](#): group-convolutional structure is necessary and sufficient for equivariance to G , determined by Peter–Weyl decomposition. Whether the resulting map is topology-faithful is a separate question depending on whether the irreducible representations of G separate points of Q , which is established for the classical compact Lie groups.

Remark 10.2 (Existence bounds are not learnability guarantees).

The bounds in Table 6 answer only the existence question: they establish the most compressed latent space in which a faithful map *can* exist, not that gradient descent will discover one. The discovery question is addressed in §12.3.

Part IV — Validation and Open Problems

Empirical evidence and open problems

11 Empirical Validation: Character-Count Manifolds

Reframing: Character-count manifolds as mechanistically recovered quotient structure. [Direct mechanistic evidence]

Gurnee et al. [2026] mechanistically identify how Claude 3.5 Haiku represents character position within a line of text: a one-dimensional variable is encoded on a low-dimensional curved manifold (a helix-like curve) in representation space, discretised by overlapping feature families analogous to biological place cells, with attention heads “twisting” the manifold to detect approaching line boundaries.

This is direct mechanistic evidence that a neural network internally recovers a continuous quotient variable — character count as a 1D quotient of raw token position — and stores it on a topologically faithful low-dimensional submanifold of the residual stream.

The recovered manifold dimension ($\dim(Q) = 1$, $\text{emb}(Q) = 2$) matches the bounds of Table 6 for a 1D quotient exactly. The observation that the quotient is stored on a 2D submanifold (a helix in $\mathbb{R}^3$) of the residual stream is consistent with the primary width floor: the effective latent dimension used to encode this quotient variable is at least 2, as Proposition 3.1(i) requires of the final representation space for $Q = S^1$. This is not a claim that the layer in question independently achieves topology-faithful encoding; it is a mechanistic observation that the network has spontaneously allocated at least the minimum necessary dimensions to represent this quotient.

Three features of this case are noteworthy for the architecture bounds programme. First, the quotient is not declared by design — it is recovered spontaneously from prediction-task training, confirming that the dimension floor is respected even in unconstrained learning. Second, the helix topology (a manifold in $\mathbb{R}^3$ with the topology of S^1 , which requires width ≥ 2 and is stored in a 3D subspace) suggests the network uses one ambient dimension beyond the embedding floor as “slack” for the discretisation structure. Third, the overlapping feature families (place-cell-like tiles covering the helix) are consistent with the overparameterisation-as-stabiliser conjecture of Farkas [2026b]: surplus width above $\text{emb}(Q)$ is used to encode the quotient structure in a perturbation-stable distributed form.

12 Open Problems

Five problems are left open by the analysis of this paper, plus one scope-limiting note on an extension developed elsewhere.

12.1 Typed Architecture Grammar

A typed compositional grammar of architecture operations — projection, convolution, attention, pooling, normalisation — classified by their effect on quotient structure, would make the framework constructive end-to-end. Each operation would carry a type signature; composing operations would compose signatures; a well-typed architecture would provably preserve the declared $\sim$. Category-theoretic foundations are available [Shiebler et al., 2021].

Three recent systems provide partial instantiations.

Remark 12.1.1 (DeepSeek Sparse Attention as a learnable admissibility declaration).

DeepSeek-AI [2026] introduce DeepSeek Sparse Attention (DSA): each query token attends only to its top- k most relevant key-value entries. In the paper’s vocabulary, DSA makes an explicit admissibility declaration at every attention step: the top- k tokens are declared semantically relevant, the remaining $L-k$ discarded. In the typed grammar framework, DSA’s type signature would be: $\text{sparse-attention}(k)$ preserves the quotient structure of the top- k token relationships while collapsing the bottom- $L-k$. Falsifiable prediction: DSA should fail to preserve topological features requiring more than k tokens to represent, testable via the simplicial-tower diagnostic of Paluzo-Hidalgo [2025] across varying k .

Remark 12.1.2 (λ -RLM as a near-realisation of typed grammar at the reasoning level).

Roy et al. [2026] (λ -RLM) introduce a typed functional runtime for long-context reasoning grounded in λ -calculus, with a library of pre-verified combinators. Neural inference is used only on bounded leaf subproblems; all control flow is handled by the typed combinator library, yielding formal guarantees: termination, closed-form cost bounds, and controlled accuracy scaling with recursion depth. This is the closest existing realisation of the typed grammar vision — at the level of reasoning operations rather than neural architecture operations. The formal machinery is the same; only the domain differs.

12.2 Diffusion Models and the Marginalised Quotient Hierarchy

The forward diffusion process at noise level σ collapses all pairs (x, x') with $\|x - x'\| \lesssim \sigma$ to the same equivalence class under $q_\sigma : \mathcal{X} \rightarrow Q_\sigma$. The noise schedule induces a nested hierarchy of quotient maps. The marginal density $p(u) = \int p(u|t) p(t) dt$ integrates over all noise levels. Sahraee-Ardakan et al. [2026] show that noise-agnostic diffusion models perform a Riemannian gradient flow on $E_{\text{marg}}(u) = -\log p(u)$, with a locally conformal metric. In the paper’s terms, the model implicitly navigates a marginalised quotient hierarchy.

A concrete instance is provided by Heek et al. [2026] (Unified Latents), who co-train an encoder, a diffusion prior, and a diffusion decoder under a single ELBO objective,

obtaining an interpretable bound on latent bitrate. Lower-bitrate latents correspond to coarser quotient resolution and are easier to model but lose reconstruction fidelity — the compression-faithfulness tradeoff of Theorem 4.4 of Farkas [2026a] at the latent-diffusion level.

Falsifiable prediction. Noise-prediction models suffer from a Jensen gap: $\|\mathbb{E}[\varepsilon|u]\|^2 \leq \mathbb{E}[\|\varepsilon|u, t\|^2]$. Velocity-based parameterisations avoid this. Testable prediction: models trained with velocity parameterisation should produce metrically less distorted representations near the data manifold than noise-prediction models, independently of topological faithfulness.

12.3 Discovery Upper Bounds: From Existence to Learnability

The Upper Bound Programme answers the *existence* question: what width suffices for a topology-faithful map to exist? The *discovery* question is distinct: under what conditions on architecture, initialisation, data distribution, and training procedure will gradient descent on a network of width $\text{emb}(Q)$ converge to a topology-faithful encoding?

The three failure modes of Farkas [2026a] already establish negative answers for broad families of objectives: any width will converge to traversal-consistent (Theorem 4.6), graded-compression (Theorem 4.4), or dimension-collapsed (Corollary 3.6) solutions without additional structural constraints. Even with those traps avoided via equivariant architecture, structural regularisation, and sufficient width — no positive result currently guarantees discovery.

The gap is not merely technical. The embedding theorems of §§5–8 are existence results proved by generic-position or algebraic construction arguments. These are non-constructive: they show that a faithful map exists without specifying which gradient-descent trajectory finds it, or whether any does. The Guliyev–Ismailov hierarchy (§6) is the closest existing result, but it operates in the approximation-theoretic setting where the target function is known, not in the learning setting where it must be inferred from data.

A discovery upper bound would take the form: *if the data distribution satisfies condition $\mathcal{D}$, the architecture satisfies condition $\mathcal{A}$, and the objective satisfies condition $\mathcal{O}$, then with probability $\geq 1 - \delta$ over initialisations and mini-batch sequences, a network of width $\text{emb}(Q)$ converges to a topology-faithful encoding of Q within T steps.* No such result currently exists for any non-trivial quotient Q beyond the contractible case. Establishing even a single instance — for $Q = S^1$ with a known periodic data-generating process — would be a significant advance.

A related open question concerns the admissibility source taxonomy of Farkas [2026b]. That paper characterises four identified sources through which admissibility can enter training (known symmetry groups, causal invariances, augmentation supervision, domain declaration), but does not claim these are exhaustive. Two candidate mechanisms remain uncharacterised. First, overparameterisation may function as an implicit admissibility source: if the prior induced by large capacity concentrates disproportionately on quotient-aligned solutions — either by the Kolmogorov complexity simplicity-bias framework of Dingle and Hutter [2026] — quotient-aligned functions factor through a lower-dimensional Q and have lower KC than traversal interpolants, so deep networks with simplicity bias [Valle-Pérez et al., 2018, Mingard et al., 2025] concentrate probability mass on them — or by the data-driven prior convergence documented in Huh et al. [2024] — then admissibility may be supplied without explicit declaration of $\sim$, and the discovery programme changes character. Second, biological evolution may constitute

an admissibility source operating at a timescale and through a fitness criterion outside the taxonomy: selection pressure acts over generations to identify the quotient structure relevant to survival, without any within-organism declaration of $\sim$. Whether either mechanism produces topology-faithful representations of a specific Q without an explicit admissibility source is not established.

12.4 Non-Integer-Dimension Quotients: From Prevalence to Determinism

The SYC theorem (§5.2) gives a prevalence sufficiency condition $m > 2d_{\text{box}}$ for compact sets of non-integer box-counting dimension. Many practical quotients have non-integer dimension: strange attractors of dynamical systems (Lorenz with $d_{\text{box}} \approx 2.062$, Rössler, Hénon, Kuramoto–Sivashinsky), Whitney-stratified spaces with strata of mixed dimension, self-similar sets arising from iterated function systems, and natural data manifolds with fractal margins. Four questions are left open.

Open Problem 4d (Embeddability at all). The SYC theorem, and all open problems that follow, assume the quotient embeds into some finite-dimensional $\mathbb{R}^m$. For non-integer-dimension sets this is a prior question, not a given. The classical embedding hierarchy (Menger–Nöbeling, Whitney, Haefliger) is stated for compact metric spaces of integer *covering* dimension. For sets whose Hausdorff or box-counting dimension is non-integer, the relation between these finer dimensions and covering dimension is non-trivial: a compact metric space of finite covering dimension always embeds in some $\mathbb{R}^m$ by Menger–Nöbeling, but box-counting and Hausdorff dimensions can exceed covering dimension and can even fail to be finite without further regularity. Whether a given non-integer-dimension quotient embeds into $\mathbb{R}^m$ at all — and, if so, what the operative dimension invariant is — is prior to the upper and lower bound questions below. The SYC bound $m > 2d_{\text{box}}$ is conditional on embeddability being given; it does not establish it.

Open Problem 4a (Deterministic lower bound). Does a deterministic embedding obstruction hold for fractal attractors? Concretely: for the Lorenz attractor with $d_{\text{box}} \approx 2.062$, does every continuous injective map $A \hookrightarrow \mathbb{R}^m$ require $m \geq 5$? The SYC theorem does not answer this. An affirmative answer would upgrade SYC’s prevalence result to a two-sided floor analogous to the smooth-manifold results of this paper.

Open Problem 4b (Learning-setting SYC). SYC assumes the embedding map is drawn randomly (in the prevalence measure). Gradient descent selects maps non-randomly: it follows the loss landscape. Can the SYC prevalence guarantee be transferred to the learning setting? That is: if $m > 2d_{\text{box}}$ and training converges, does convergence happen to an injective encoder with high probability over random initialisation? This requires understanding whether the set of non-injective encoders is a gradient trap, not merely a measure-zero set.

Open Problem 4c (Stratified spaces). Many practical quotients are Whitney stratified spaces: piecewise smooth with strata of varying dimension. The Lorenz attractor is an example: smooth wings (dimension 2) meeting along a one-dimensional stable manifold. The effective dimension for the SYC bound is unclear for stratified sets. Extending the floor analysis to Whitney stratified quotients would close the gap between the smooth case (§5) and the fully fractal case covered by SYC.

Remark 12.3.1 (A non-gradient positive result: transversal geometry on

ARC).

The learnability gap concerns gradient-based learning. A striking existence result in a different paradigm is provided by Khalil [2026], whose transversal ARC solver discovers the correct output transformation for grid tasks with zero gradient descent. The solver represents cell-pair relationships as Plücker lines in projective 3-space P^3 , then recovers the transformation rule by finding *transversals* — lines simultaneously meeting multiple training lines, whose existence (exactly 0 or 2 for 4 lines in general position) is guaranteed by Schubert calculus. Of non-timeout tasks, the solver achieves 86% on the ARC training set and 95% on the evaluation set.

The connection to this paper’s framework is precise: the transversal solver recovers the quotient map by geometric intersection rather than task-loss minimisation, and is immune to all three failure modes by construction. Its success confirms that the quotient structure is real and discoverable, while demonstrating that the standard neural mechanism is not necessary. The open question for the neural setting is whether the geometric constraints the solver exploits explicitly can be embedded into a gradient-based learning procedure.

12.5 Single-Quotient Scope

All floor theorems in this paper are stated for a single global quotient Q . The generalisation to problems with regionally varying quotient structure — a *quotient atlas* — is developed in Paper 4 (forthcoming). We note honestly that this extension only became visible after examining concrete problems (the double pendulum of Paper VII §7 and the Color Fantasy ship hotel system of Paper VII §8); the general framework does not appear from the single-quotient theory alone. Developing the quotient-atlas floor theorems is therefore scoped to Paper 4 rather than retrofitted here.

12.6 Operative Contract for Quotient-Faithful Learning

The lower-bound programme of Part II proves the *topological* floor $m \geq \text{emb}(Q)$ (Proposition 3.1). §2.4 establishes that stricter contracts — smooth, bi-Lipschitz, isometric — impose strictly larger floors (Proposition 2.4, Theorem 2.4). The open question is which contract is the operative floor for quotient-faithful learning with a given objective and architecture.

A metric-preserving floor (bi-Lipschitz with controlled distortion) is likely the right contract for most downstream tasks, but deriving it requires results beyond classical embedding theory: the contract must be stable under gradient updates, realizable within the parameterised function family of the architecture, and consistent with the loss function’s sensitivity structure. Cross-entropy with softmax is sensitive to angular structure but not scale; contrastive objectives are explicitly metric; regression is Lipschitz-sensitive at a quantitative level set by the output loss. In each case the relevant floor is likely above the topological one, but by how much depends on the objective. Establishing the operative contract for even a single canonical objective/architecture pair would be a significant advance.

A Definitions and Key Concepts

This appendix collects the formal definitions from the main text.

Inherited from Paper I [Farkas, 2026a].

Def. 2.1. Admissible nuisance relation. Equivalence relation $\sim$ on $\mathcal{X}$ declaring semantically irrelevant variations.

Def. 2.2. Lawful quotient, encoder, topology-faithful encoding. (Q, q) the semantic quotient; $E : \mathcal{X} \rightarrow Z$ the encoder; $\tilde{E}$ topology-faithful if a homeomorphism onto its image.

New in Paper 3.

Def. 4.1. Graded quotient filtration. $r_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ min-transitive; semantic ultrapseudometric $d_c = 1 - r_c$; nested filtration $\{Q_{c,\varepsilon}\}$. (Citing Paper 2.)

Def. 7.3.1. Federer reach. $\tau(\phi(Q))$: largest r such that every point within distance r from $\phi(Q)$ has a unique nearest point on $\phi(Q)$. Controls both local (Jacobian) and global (distant strands) approximation failure.

New in Paper 3, §2.4 (stronger contracts).

Def. 3.4.1. Bi-Lipschitz embedding dimension at distortion D . $\text{emb}_{\text{bi-Lip}}(Q, D) := \min\{m : Q \text{ admits a bi-Lipschitz embedding into } \mathbb{R}^m \text{ with distortion } \leq D\}$, with the convention that this is ∞ if no such embedding exists for any finite m . Reduces to $\text{emb}(Q)$ as $D \rightarrow \infty$.

Def. 3.4.2. Contract hierarchy. The strict hierarchy $\text{topological} \subset C^\infty \text{ smooth} \subset \text{bi-Lipschitz} \subset C^1 \text{ isometric} \subset C^\infty \text{ isometric}$ on admissible induced maps $\tilde{E} : Q \rightarrow Z$, each imposing a potentially stricter dimension floor than the one below.

Architecture floor.

Width floor (Prop. 3.1). Any topology-faithful encoding $\tilde{E} : Q \rightarrow Z$ requires $\dim(Z) \geq \text{emb}(Q)$. A non-bypassable layer ℓ must also have width $\geq \text{emb}(Q)$.

Scale-indexed floor (Prop. 4.2). Topology-faithful encoding of $Q_{c,\varepsilon}$ requires latent width $\geq \text{emb}(Q_{c,\varepsilon})$, non-decreasing as $\varepsilon \rightarrow 0$.

Bi-Lipschitz floor (Prop. 3.2). A bi-Lipschitz induced map $\tilde{E} : Q \rightarrow Z \subseteq \mathbb{R}^m$ with distortion $\leq D$ requires $m \geq \text{emb}_{\text{bi-Lip}}(Q, D) \geq \text{emb}(Q)$. For Q with doubling dimension δ , the floor is controlled by δ up to constants (Naor–Neiman 2012).

Smooth/isometric floors (Thm. 3.3). Whitney, Nash–Kuiper, and Nash–Günther classical floors apply for C^∞ -smooth, C^1 -isometric, and C^∞ -isometric contracts respectively.

B Proofs and Proof Sketches

B.1 Results Inherited from Farkas (2026a)

All formal results restated in §2 are proved in Farkas [2026a], Appendix A. This paper adds no new proofs of those results.

Result	Classical source	Paper	1 location
Theorem 4.6 (Traversal Ambiguity)	Whyburn 1942; Rourke–Sanderson 1972	Paper App. A.2	1,
Theorem 4.4 (Task-Map Collapse)	Lehmann–Scheffé 1950 (i–ii); novel (iii)	Paper App. A.3	1,
Lemma 3.4 (Quotient Embedding Requirement)	Menger 1932; Kvalheim–Sontag 2024	Paper App. A.4	1,
Theorem 3.3 (Covering-Dimension Floor) and Corollary 3.6 (Below-Floor Collapse)	Brouwer 1911; Hurewicz–Wallman 1941	Paper App. A.5	1,
Corollary 3.6 (Below-Embedding Collapse)	Contrapositive of Thm. 7.4	Paper App. A.6	1,

B.2 Proof of Corollary 3.2 (Product Decomposition Lower Bound)

By the product dimension formula for compact metric spaces [Engelking, 1978, Theorem 1.5.14], $\dim(Q_1 \times \cdots \times Q_k) = \sum_i \dim(Q_i)$. Theorem 3.3 of Farkas [2026a] then gives $m \geq \dim(Q) = \sum_i \dim(Q_i)$. The formula holds when each Q_i is a compact metric space, covering all process-generated quotient classes of §3.2. $\square$

B.3 Results Cited in the Upper Bound Programme (§§5–9)

None of the following proofs are new contributions of this paper; all are cited applications of established mathematics.

Menger–Nöbeling [Menger, 1932, Nöbeling, 1931]. Classical (1931–32). Every separable metrizable space of covering dimension $\leq n$ embeds in $\mathbb{R}^{2n+1}$. Proof uses the Baire category theorem to show homeomorphic embeddings form a dense G_δ subset of $C(X, \mathbb{R}^{2n+1})$. Tightness: the n -skeleton of the $(2n+2)$ -simplex embeds in $\mathbb{R}^{2n+1}$ but not in $\mathbb{R}^{2n}$ [Flores, 1935, van Kampen, 1932].

Whitney [Whitney, 1944]. Classical (1944). Every smooth n -manifold embeds in $\mathbb{R}^{2n}$. The Whitney trick eliminates double points of a generic immersion $M^n \looparrowright \mathbb{R}^{2n}$ by constructing embedded Whitney discs cancelling pairs of transverse intersection points with opposite sign. Tightness for $n = 2^k$: Stiefel–Whitney obstruction.

Haefliger [Haefliger, 1961]. Classical (1961). k -connected compact M^n with $n \geq 2k+3$ embeds in $\mathbb{R}^{2n-k}$. Condition $n \geq 2k+3$ is sharp: Haefliger’s own example of S^3 knotted in S^6 shows necessity.

Stiefel–Whitney lower bounds. If M^n embeds in $\mathbb{R}^{n+c}$, Whitney duality forces $\bar{w}_i(M) = 0$ for $i > c$. For $\mathbb{R}P^2$: $w(T\mathbb{R}P^2) = (1+a)^3 = 1+a+a^2$ over $\mathbb{Z}/2$; $\bar{w} = 1+a$; $\bar{w}_1 = a \neq 0$ forces $\text{emb} \geq 3$; non-orientability forces $\text{emb} \geq 4$. For $SO(3) \cong \mathbb{R}P^3$: all $\bar{w}_i = 0$ (parallelisable); Hopf 1940 linking-number argument forces $\text{emb} \geq 5$; Wall gives $\text{emb} \leq 5$. For $\mathbb{C}P^2$: the SW argument gives only $\text{emb} \geq 6$ ($\bar{w}_2 = h \neq 0 \Rightarrow \text{codim} \geq 2$); the tight lower bound $\text{emb} \geq 7$ requires the signature obstruction $\sigma(\mathbb{C}P^2) = 1 \neq 0$, as detailed in the main text above.

Wall’s theorem [Wall, 1965]. Classical (1965). Every compact connected smooth 3-manifold embeds in $\mathbb{R}^5$. All closed orientable 3-manifolds are parallelisable [Stiefel,

1936] ($w_1 = w_2 = 0$); a general-position argument in $\mathbb{R}^5$ via handle decomposition then produces an embedding.

Guliyev–Ismailov constructive results [Guliyev and Ismailov, 2016, 2017, 2021]. All three theorems are proved in the cited papers. The single-neuron construction (G–I 2016) uses the Calkin–Wilf enumeration of rationals to construct a C^∞ sigmoidal activation representing a dense family of polynomials. The two-neuron fixed-weight result (G–I 2017) is constructive for the positive direction and uses a counting/linear-independence argument for the negative direction. The TLFN with $3d + 2$ neurons (G–I 2021) implements the Kolmogorov–Arnold representation with unit-coordinate weights.

Hirsch’s theorem [Hirsch, 1976]. Classical (1976). $\text{Emb}^k(M, N)$ is open in $C^k(M, N)$ for $k \geq 1$. Proof: the inverse function theorem and compactness give a lower bound on the minimum singular value of Jf ; an explicit ε -neighbourhood in C^1 topology consists entirely of injective maps.

Federer’s reach [Federer, 1959]. Classical (1959). Simultaneously bounds local (Jacobian degeneration) and global (distant strands) sources of approximation failure. Allowable perturbation $\varepsilon^* \lesssim \tau/C$ for universal C .

Niyogi–Smale–Weinberger [Niyogi et al., 2008]. Classical (2008). Under Hausdorff proximity $d_H(A, \phi(Q)) < \sqrt{3/5} \tau$, the nearest-point projection from A to $\phi(Q)$ is unique and provides an explicit deformation retraction.

Naor–Neiman Assouad-type embedding [Naor and Neiman, 2012]. For every doubling constant K there exists $N = N(K)$ such that every separable metric space (X, d) with doubling constant $\leq K$ admits a bi-Lipschitz embedding of its $(1 - \varepsilon)$ -snowflake into $\mathbb{R}^N$ with distortion $D(K, \varepsilon)$. The ambient dimension N depends only on the doubling constant, not on the snowflake exponent. Used in Paper III §2.4 as the upper-bound side of Proposition 2.4; the matching lower bound $\text{emb}_{\text{bi-Lip}}(Q, D) \gtrsim \delta$ is the classical volume-comparison argument.

Nash–Kuiper C^1 isometric embedding [Nash, 1954, Kuiper, 1955]. Classical (1954–55). Any short C^∞ embedding $Q^n \hookrightarrow \mathbb{R}^m$ with $m \geq n + 1$ can be C^0 -approximated by a C^1 isometric embedding. The conclusion is counter-intuitive — isometric embedding of n -manifolds in just $n + 1$ dimensions at C^1 regularity — because the embedding is C^1 but not C^2 . Nash’s construction uses successive perturbations along orthogonal directions; Kuiper’s simplifications yield the cleanest proof.

Nash–Günther C^∞ isometric embedding [Nash, 1956, Günther, 1991]. Classical (1956, 1991). Every compact Riemannian n -manifold admits a C^∞ isometric embedding into $\mathbb{R}^m$ for $m \geq \max(n + 5, (n + 2)(n + 5)/2)$. Nash’s original proof uses a hard-implicit-function (Nash–Moser) technique; Günther’s 1991 proof replaces this with a standard fixed-point argument on an elliptic PDE, giving a much shorter proof of essentially the same bound. The dimension $m = O(n^2)$ is quadratically worse than the C^1 isometric case and reflects genuine rigidity of C^∞ isometric embeddings.

Proof of Proposition 9.2.1 (Additive upper bound). Let $\phi_i : Q_i \hookrightarrow \mathbb{R}^{m_i}$ be homeomorphic embeddings. Define $\Phi = (\phi_1, \dots, \phi_k)$. Each ϕ_i is injective, so Φ is

injective (if $\Phi(x) = \Phi(y)$ then $\phi_i(x_i) = \phi_i(y_i)$ for each i). As a continuous injection from a compact space to a Hausdorff space, Φ is a homeomorphism onto its image. $\square$

Proof of the tight sphere-product theorem. *Upper bound:* Induction. For $k = 1$: $S^{n_1} \subset \mathbb{R}^{n_1+1}$. Inductive step: given $S^{n_1} \times \dots \times S^{n_{k-1}} \subset \mathbb{R}^{1+\sum_{i < k} n_i}$, revolve around a new coordinate hyperplane. For $S^{n_1} \times S^{n_2}$: $(x, y) \mapsto ((2 + y_1) \cdot x_1, \dots, (2 + y_1) \cdot x_{n_1+1}, y_2, \dots, y_{n_2+1})$ embeds $S^{n_1} \times S^{n_2}$ in $\mathbb{R}^{n_1+n_2+1}$. *Lower bound:* The product is a closed $(\sum n_i)$ -manifold; invariance of domain gives $\text{emb} \geq 1 + \sum n_i$. Bounds match. $\square$

C Contribution Attribution and AI Disclosure

C.1 Context

This paper is the third in a series on quotient faithfulness in representation learning. The attribution framework follows Farkas [2026a] and Farkas [2026b]: an AI-assisted writing process (mathematical interlocution, prose drafting, and LaTeX construction involving Claude Sonnet 4.6 / Anthropic, ChatGPT 5.4 / OpenAI, Gemini 3 / Google, and MetaAI / Meta, used in varying roles across sessions) with a ++/+/-0 contribution system. Under this system, ++ denotes an *unsolicited* AI follow-up contribution by any of the tools used; + denotes an AI contribution made in response to author direction; 0 denotes a contribution not involving AI tools. The AI column records the maximum weight across all AI tools that contributed to a given row.

C.2 Architecture-Relevant Contribution Weight

C.3 Primary Contribution

The primary contribution of this paper is the synthesis of the upper bound programme for process-generated quotients: connecting the classical embedding hierarchy (Menger–Nöbeling, Whitney, Haefliger), the Guliyev–Ismailov constructive approximation results, the Federer–NSW quantitative C^1 bridge, and the Thurston–Perelman finite classification into a single coherent architectural framework with a practical lookup table. Each individual component is classical; the synthesis and its application to representation learning architecture design are original.

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- Peter Farkas. Quotient atlases in representation learning: Region-varying quotient structure and its dimension floor. Preprint, 2026d. Paper IV in the series *Quotient Faithfulness in Representation Learning*. Extends the dimension-floor programme to region-varying quotient structure via quotient atlases. Develops the given-partition, learned-partition, and emergent-discovery tiers. Proves the partition base-split atlas dimension-floor theorem. Åbo Akademi University, DAZE Project.
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D Attribution Methodology

The attribution methodology (the $++/+ / 0$ weighting scheme and its worked example) is documented in full in Paper I [Farkas, 2026a], Appendix D, and Paper 2 [Farkas, 2026b], Appendix D. The same scheme is applied in this paper; readers seeking the full prompt and methodology should consult those appendices.

Result / contribution	Character	Lit.	Author	AI
Thms. 7.2 and 7.4; Cor. 7.5 (from Paper 1)	inherited	++	0	0
Architectural framing of Cor. 7.5 (Prop. 3.1)	observation	0	++	0
Scale-indexed floor (Prop. 4.2)	definitional consequence	0	++	0
Embedding hierarchy synthesis (§5)	synthesis	++	+	+
Stiefel–Whitney lower bounds	synthesis	++	+	++
G–I hierarchy to 1D quotient encoding (§6)	observation	++	++	+
C^1 bridge (Hirsch + Federer + NSW) (§7)	synthesis	++	++	++
NSW $\sqrt{3/5}\tau$ quantitative threshold	observation	++	0	++
Thurston geometry table with Wall bounds (§8)	synthesis	++	+	+
Poincaré/decidability architecture implication	observation	+	++	0
Physics motivation ($Q_{\text{space}} \times Q_{\text{time}}$)	observation	+	++	0
Additive upper bound + tight sphere-product theorem (§9)	classical application	++	+	++
Product composition rules	synthesis	++	+	+
Architecture lookup table (Tab. 6)	synthesis	0	++	+
Remark 10.2: existence vs. learnability distinction	observation	0	++	0
Character-count reframing (§11)	novel reframing	+	++	0
§12.3: Discovery upper bounds open problem	observation	0	++	0
LaTeX construction and file structure	—	0	0	++
KC infeasibility note in Remark 3.1.2	connecting note	++	+	+
Dingle–H. formal citation in §12.3 overparam. remark	connecting note	++	+	+
Notation page, definitions appendix, structural additions	structural	0	+	++
Architecture-level reframing of Secondary Claim (§2.3)	reframing	0	+	+
“Width is the only continuous capacity” three observations (§2.3)	correction cycle	+	+	+
Terminology: “non-bypassable layer”	terminological	0	+	++
Bi-Lipschitz floor (Prop. 2.4)	classical packaging	++	+	+
Contract hierarchy & classical isometric floors (§2.4, Thm. 2.4)	survey synthesis	++	+	+
Open Problem: Contract selection (§12.6)	observation	0	0	++
Figure 1 intro cleanup and caption fixes	editorial	0	+	+

Table 7: Architecture-relevant contribution attribution. Columns: **Lit.** = prior literature; **Author** = Peter Farkas; **AI** = the AI tools used in drafting this paper (Claude Sonnet 4.6 / Anthropic, ChatGPT 5.4 / OpenAI, Gemini 3 / Google, MetaAI / Meta). Weight key: ++ = primary responsibility; + = substantive contribution; 0 = not applicable.